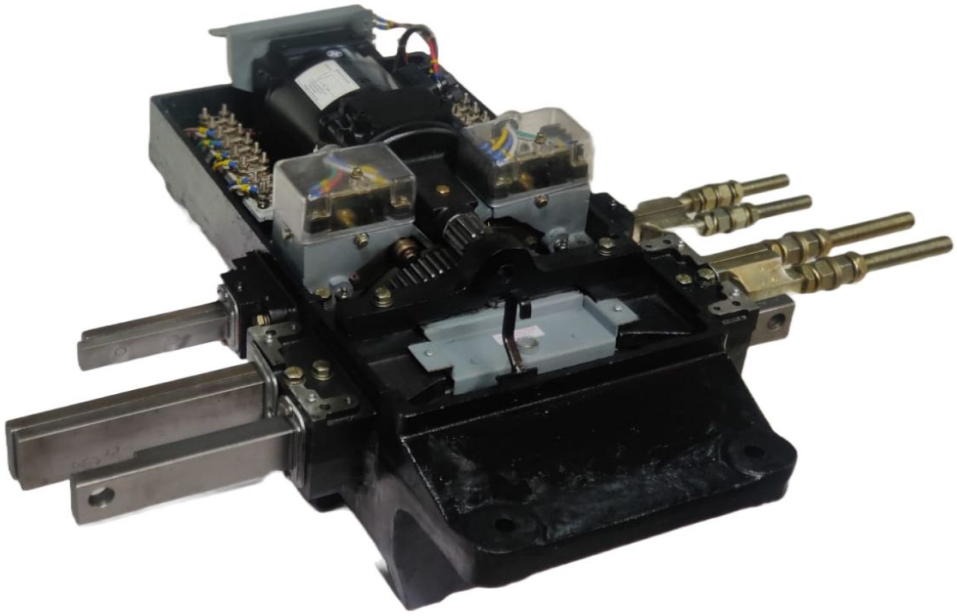
EXCELLENCE IN MAINTENANCE

भारत सरकार - GOVERNMENT OF INDIA

रेल मंत्रालय - MINISTRY OF RAILWAYS

केवल कार्यालयीन प्रयोग हेतु
For Official Use only

HANDBOOK ON
GUIDELINES FOR MINIMIZING FAILURES OF ELECTRIC POINT MACHINE



CAMTECH/S&T/PROJ/2024-25/EPM/Ver.1.0
March-2025

END User: SSE/JEs & Maintenance Staff of S&T dept. of Indian Railway

INDIAN RAILWAYS CENTRE FOR ADVANCED MAINTENANCE TECHNOLOGY
Maharajpur, Gwalior (M.P.) – 474005

Blank Page

Handbook on Guidelines for Minimizing Failures of Electric Point Machine

Prepared by: *Jitendra Singh Gour, SSE/Sig/CAMTECH*

(Under the Guidance of Shri Yogendra Pratap Singh, Director/S&T/CAMTECH)

FOREWORD

Foreword

The Handbook on Guidelines for Minimizing Failures of Electric Point Machines is an essential reference for engineers and maintenance staff of S&T department of Indian railways. This handbook gives an overview about various parts and working of point machine and also provides a comprehensive set of maintenance instructions, best practices, and troubleshooting techniques to prevent or minimize their failures.

Failures of Electric Point Machines can stem from a variety of factors, including mechanical wear and tear, electrical faults, environmental conditions, improper maintenance, or even design flaws. The impact of these failures can be far-reaching, potentially causing delays, safety hazards, and increased operational costs.

The guidelines offered within this publication serve as a practical tool for railway professionals to develop more effective maintenance schedules, diagnosing malfunction in machine if any, and ultimately reduce the occurrence of point machine failures in the field.

I hope that the instructions given in this handbook will be quite helpful to the S&T personnel associated with various types of Construction Projects and as well as Maintenance of Signaling assets in Indian Railways and taking preventive measures to avoid safety hazards.

CAMTECH, Gwalior ***Dinesh Shukla***
Date: March 2025 ***Principal Executive Director***

PREFACE



Yogendra Pratap Singh, IRSSE 2009
Director Signal /CAMTECH & RDSO LKO
Experience: 13 Years

Commissioned more than 50 stations in South Central Railway (SCR), which includes the first third line of SCR, doubling projects in Nanded Division, various new line works, replacement work of EI/PI/IB, commissioning of new IB and ABS systems, LC gate interlocking, tunnel communication in Margutti Tunnel of SC Division and several alterations works.

****

The **Handbook on Guidelines for Minimizing Failures of Electric Point Machines** is a comprehensive resource aimed at providing railway engineers, maintenance personnel, and operators with essential practices and strategies to enhance the reliability and longevity of electric point machines used in signalling and track switching systems.

Electric Point Machines play a critical role in ensuring the efficiency and smooth operation of railway networks. However, as with all complex mechanical and electrical systems, they are susceptible to failures that can disrupt services and pose risks to safety.

This handbook serves as a proactive guide to preventing these failures by addressing common causes, providing detailed maintenance practices, and offering troubleshooting techniques. Through this publication, we seek to empower professionals with the knowledge needed to diagnose failures, to adopt proper maintenance routines, and to stay up-to-date with the latest technological advancements in electric point machine systems.

We are sincerely thankful to Signal Directorate RDSO Lucknow , MDZTI / S.C.Rly / Secunderabad ,Signal Workshop, N.E. Rly /Gorakhpur, Sr. D.S.T.E./Co., Western Railway , Mumbai Central, M/s Globe Scott Motors (P) Ltd., Valsad and S&T personnel of Indian Railways who helped us in preparing this handbook by providing valuable information

CAMTECH, Gwalior ***Yogendra Pratap Singh***

Date: March 2025 ***Director(S&T)/CAMTECH***

TABLE OF CONTENTS

INDIAN RAILWAYS CENTRE FOR ADVANCED MAINTENANCE TECHNOLOGY 1

FOREWORD i

PREFACE ii

TABLE OF CONTENTS iii

OUR OBJECTIVE vi

DISCLAIMER vii

1.1 INTRODUCTION: 1

1.2 MAIN PARTS OF ELECTRIC POINT MACHINE: 1

1.3 BRIEF OVER VIEW OF IMPORTANT PARTS OF ELECTRIC POINT MACHINE: 2

1. 4 GROUND CONNECTIONS : 12

1. 5 LIST OF INSULATION MATERIAL FOR POINT TURNOUTS IN TRACK CIRCUIT AREA : 14

2. 1 INSTALLATION OF ELECTRIC POINT MACHINE (143mm): 16

2.1.1. Requirements: Electric Point Machines of approved type shall be used. 16

2.1.2 Point Machine, Notches, Gaskets, Sealing: 17

2.1.3 Adjustment of Driving Rod 17

2.1.4 Adjustment of Detector Contacts: 17

2.1.5 Adjustment of Friction Clutch: 18

2.1.6 Testing of Point Machine :→ 18

2.2 ELECTRIC POINT MACHINE 220 mm THROW WITH CLAMP TYPE LOCK FOR THICK WEB SWITCHES (TWS): 19

2.2.1 Conventional over-riding type of switches suffers from geometrical deficiencies like: 19

2.2.2 Benefits of Thick Web Switches (TWS) over conventional over-riding type of 20

Switches: 20

Thick Web Switches (TWS) overcome deficiencies of conventional over-riding type of 20

Switches. These are non-over riding type : 20

2.2.3 For ground connection and clamp lock as per RDSO Drg. No. RDSO/S-3454 for 60 Kg 20

rail and RDSO/S-3455 for 52 Kg rail, IRS point machine of 220 mm throw is used. 20

2.3 OPERATING PARAMETERS OF EPM (143mm AND 220mm) AS PER IRS: S-24-2025 21

REV.3. : 21

3. 1 REQUIREMENTS FOR MAINTENANCE OF POINT AND POINT MACHINE: 22

3. 2 PERIODICAL MAINTENANCE OF POINT MACHINE : 23

3. 3 MAINTENANCE ITEMS TO BE CHECKED DURING JOINT INSPECTION OF POINTS 25

& CROSSINGS WITH ENGG. DEPARTMENT 25

4. 1 CHECKLIST FOR TWS (THICK WEB SWITCH) INSERTION : 34

4. 2 CHECKLIST FOR TWS (THICK WEB SWITCH) Maintenance: 36

4. 3 MAINTENANCE SCHEDULE OF ELECTRICALLY OPERATED POINTS (INCLUSIVE OF CLAMP TYPE POINT) 37

5. 1 COMMON FAILURES OF ELECTRIC POINT MACHINE (143mm):…….416

5.1.1. Mechanical failures. 42

5.1.2 Electrical failures 42

5.1.3 Other factores 42

5.2 EFFORTS TO MITIGATE FAILURES IN ELECTRIC POINT MACHINE: 43

5.2.1 Regular Maintenance and inspection: 43

5.2.2 Condition based Monitoring 43

5.2.3 Protection against environmental factors 43

5.2.4 Upgrading Infrastructure : 45

5.2.5 Power Supply Reliability : 45

5.2.6 Learning from failures: 46

REFERENCES 47

ISSUE OF CORRECTION SLIPS

The correction slips to be issued in future for this report will be numbered as follows:

CAMTECH/S&T/PROJ/2024-25/Point Machine/Ver.1.0 XX date .....

Where "XX" is the serial number of the concerned correction slip (starting from 01 onwards).

CORRECTION SLIPS ISSUED

S. No. of Correction Slip	Date of issue	Page no. and Item No. modified	Remarks

OUR OBJECTIVE

To upgrade Maintenance Technologies and Methodologies and achieve improvement in Productivity and Performance of all Railway assets and manpower which inter-alia would cover Reliability, Availability and Utilization.

If you have any suggestions & any specific comments, please write to us:

Contact Person : Director (Signal & Telecommunication)

Postal Address : Centre for Advanced Maintenance Technology,
Maharajpur, Gwalior (M.P.)
Pin Code – 474 005

Phone : 0751-2470185

Fax : 0751-2470841

E-mail : dirsntcamtech@gmail.com

DISCLAIMER

It is clarified that the information given in this handbook does not supersede any existing provisions laid down in the Signal Engineering Manual, Railway Board and RDSO publications. This document is not statuary and instructions given are for the purpose of guidance only. If at any point contradiction is observed, then Signal Engineering Manual, Telecom Engineering Manual, Railway Board/RDSO guidelines may be referred or prevalent Zonal Railways instructions may be followed. Our Objective To upgrade Maintenance Technologies and Methodologies and achieve improvement in Productivity and Performance of all Railway assets and manpower which inter-alia would cover Reliability, Availability and Utilisation. If you have any suggestion & any specific comments, please write to us: ***Contact person :*** *Director (Signal & Telecommunication)* ***Postal Address :*** *Centre for Advanced Maintenance Technology, Opposite Hotel Adityaz, Near DD Nagar, Maharajpur, Gwalior (M.P.) Pin Code – 474 005 Phone : 0751 - 2470185 Fax : 0751 – 2470841 Email : dirsntcamtech@gmail.com*

CHAPTER-1

OVERVIEW OF ELECTRIC POINT MACHINE

1.1 INTRODUCTION:

Electric Point Machine: Electric Point Machine is an electrically driven machine used for operation of Points in railway yards. It can perform the function of unlocking and operating the point switches in the desired position and lock them and detect their correct setting with the aid of an electric motor. On Indian Railways following **TWO** types of Electric Point Machines are generally used:

(i) IRS type Electric Point Machine **143 mm** stroke (Rotary Type).
(ii) IRS type Electric Point Machine **220 mm** stroke for Thick Web Switch.

Electric Point Machine with IP 67 Protections: It is water and dust proof electrically driven machine used for operation of Points in railway yards which are prone for water logging.
The following types of EPM with IP 67 Protections are being used in Indian Railway;

(i) IRS type Electric Point Machine with **IP 67 Protection 143 mm** stroke (Rotary Type).
(ii) IRS type Electric Point Machine with **IP 67 Protection 220 mm** stroke for Thick Web Switch.

Note: *The above Electric point machines* Machines conform to Specification No. IRS: S24-2025 Rev.3.0

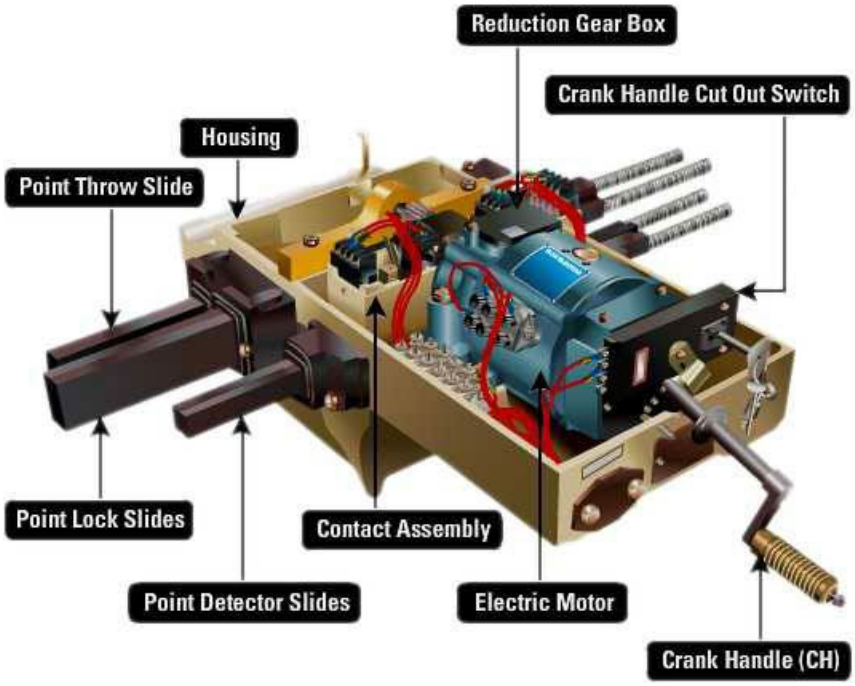


Fig.1: Electric point machine with open Cover

1.2 MAIN PARTS OF ELECTRIC POINT MACHINE:

Both types of point machines **143 mm & 220 mm** Stroke consist of following main parts:

(a) DC series split field Motor within built reduction gear unit.
(b) Gear Wheel Transmission assembly consisting of :

- Main Gear Rim
- Spring loaded friction clutch
- Transmission shaft

- Pinion
- Rotary type locking segment
- Drive disc
- Lift out disc
- Control disc

(c) Throw rod, lock slides and detection slides.
(d) Detection and control contact switch Assembly.
(e) Cast iron Case with cover.
(f) Crank handle.

The 220mm stroke Electric point machine is used for Thick Web Switch provided with additional arrangement of locking to lock the point in correctly set position termed as "Clamp type lock". The Clamp type lock consists of following components:

- Locking bar (L.H.)
- Locking bar (R.H)
- Insulating plates
- Locking washers
- Locking arms
- Stopper
- Drive lug

1.3 BRIEF OVER VIEW OF IMPORTANT PARTS OF ELECTRIC POINT MACHINE:

A) EXTERNAL PARTS :

1. Housing/Base :

- Accommodates all theparts
- Made of cast iron
- Designed to fix all parts at required place.
- Provide access to/slots for different lock and detection slides.
- Facilitate fixing of machine on sleepers
- Provides an arrangement to drain out water/oil in bottom of casting.
- Provides cable entry.



Fig.- Base of EPM

2. Cover with locking arrangement :

- Made of thick iron sheet
- Contain S&T maintainer's lock
- Secures all the parts of machine in weather proof environment
- Prevents un-authorized interference.
- Provides access only to nominated (ward & feather) crank handle insertion.
- All the apertures for keys and crank handle are secured by flaps
- All flaps has a locking arrangement



Fig.- Cover of EPM

3. Crank Handle :

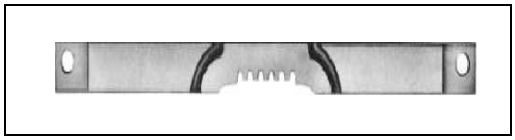
- Used for manual operation of point machine
- "Z" shape structure made of steel pipe and has wooden handle on one end to have better grip.
- Wards & feathers are welded on other end of pipe
- Facilitates coding (Allow insertion of only nominated crank handle).
- A stud is intrude in the pipe, it get engage in forked shaft of motor



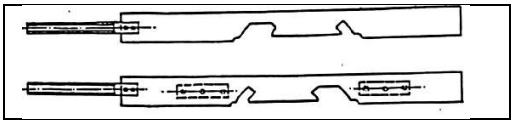
Fig.- Crank Handle

4. Slides (Total 5 Nos.)

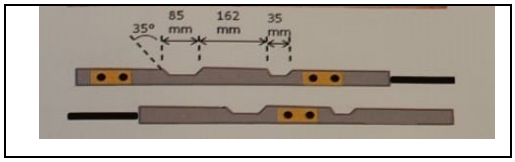
- **Throw/Drive Slide** 1 No.



- **Normal Lock slide** 1 No.
- **Reverse Lock slide** 1 No.



- **Normal Detection slide** 1 No.
- **Reverse Detection slide** 1 No.



- Throw slide is used to throw the point in normal to reverse or vice versa Direction
- Locking slides are used for locking point in Norma or Reverse positions.
- Detection slides are used to detect the point in Normal or Reverse setting.

B) INTERNAL PARTS :

1. Bearing Pedestal (02 Nos.)

- Made of Cast Iron
- Used to hold the axle of Transmission Assembly / friction clutch to the housing
- Gets damaged first when point trail through due to wrong train movement thus protecting damage to other parts of point machine

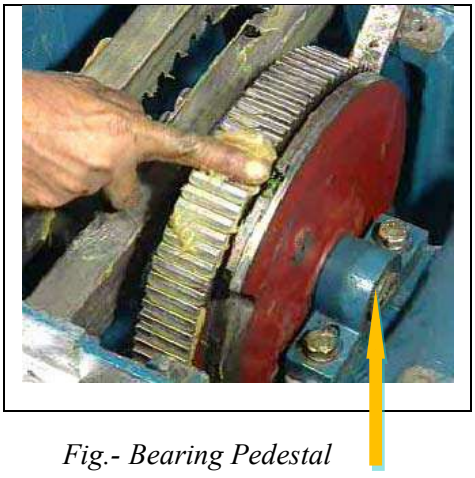


Fig.- Bearing Pedestal

2. Throw and Lock assembly:

- Part of the transmission assembly
- Used for Rotary to linear conversion mechanism
- Locking sector is used to lock the point by entering into the lock slide notches



Fig.- Throw and Lock assembly

3. Transmission assembly:

Parts of Transmission assembly are described below :

i) Drive disc (Part of Transmission Assembly)

- Extend movement to shaft hence rotate pinion and lock pawl.
- Restrict rotary movement to 270° and provide mechanical (dynamic) snubbing
- Hold gear rim on outer periphery
- Convert rotary movement into linear movement



Fig.- Drive Disc

ii) Control disc (Part of Transmission assembly)

- It is a circular disc with notch on outer periphery
- Edges of notch are vertical
- Function of control disc :
  a) Allow the top roller (roller A) to fall in side respective notch at end friction clutch rotation.
  b) Prevent upward movement of top roller (lock the top roller **)**



Fig.- Control Disc Outer view



Fig.- Control Disc Inner view

iii) Lift out disc (Part of Transmission assembly) :

- It is first disc to rotate in friction clutch.
- Function:
  a) To lift the top roller (roller A) to open detection contacts at beginning of point operation



Fig. - Lift out disc (Inner View)



Fig. - Lift out disc (Outer View)

iv) Friction clutch (Part of Transmission assembly) :

- Placed between motor and Load.
- Functions of friction clutch :
  a) To protect the motor from overload
  b) To achieve mechanical (dynamic) snubbing
  c) To follow the sequence of point operation.



Fig. - Friction Clutch Assembly



Fig. - Friction Clutch (Internal view)

Friction clutch-Features -

- Extend stroke (force) from motor to load (point),if load is within specified limit (700Kg)
- De-clutch (separate) motor and Load (point) if load is more than specified limit (700Kg).
- De-clutching of friction clutch can be adjusted to suite specified limit of load (700Kg) by loosening or tightening spring at workshops only not at site.

Friction clutch requires an adjustment if -

- Slipping current /obstruction current is more than 1.5 to 2 times of normal working current.
- Difference between normal working current and Slipping current and Obstruction current is less than 0.5 Amps.

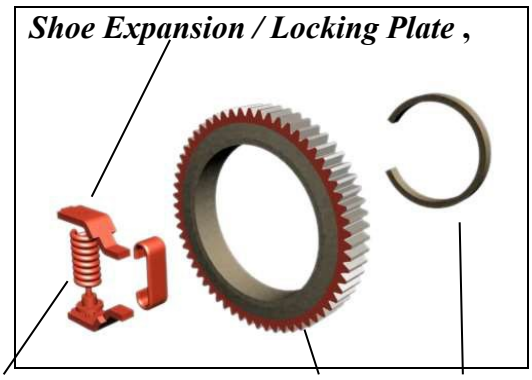


Fig.- Parts of a Friction clutch

4. Reduction Gear assembly / Box :

- Main Part of DC Motor which carry rotary motion of motor to the transmission assembly of point machine
- Appropriate reduction Gear ratio is used to get required mechanical thrust
- Motor rotate at very high speed and hence load cannot be directly put on motor shaft. Reduction ratio provides a torque-speed conversion from a higher speed motor to a slower but more forceful output.
- Reduction ratio - **20.8 : 1**
- Reduction Ratio = $\frac{\text{Number of teeth on driver}}{\text{Number of teeth on drive}}$



Fig.- Reduction Gear Assembly

5. Contactor Unit / Contact Assembly With Spring Loaded Rollers

- Comprises of two spring loaded contact groups which are fixed inside the casting.
- Function of the contact assembly:- To make and break the sets of contact.
- One group Contains two sets of contact fixed side by side
- Contact controls feed (Power) to motor and point indication relay
- Cover of contactor unit should be sealed after maintenance

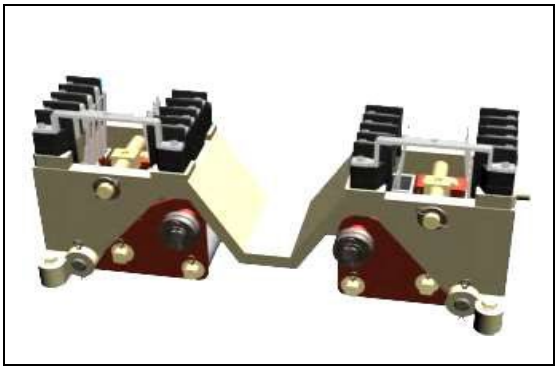


Fig.- Contactor Unit

- **Making of contacts** depend upon position of top and bottom roller on the periphery of control disc and detection slide respectively.
- Detection contact in one group & control contact in other
  Group makes when -
  a) Top roller fall in nominated notch on friction clutch
  b) Bottom roller fall in nominated notch on periphery detection slide



Fig.- Top Roller (inside notch)



Fig.- Bottom Roller (inside notch)

Note :

- Falling of top roller in notch proves correct locking of point
- Falling of bottom roller in notch proves correct setting of switches / points

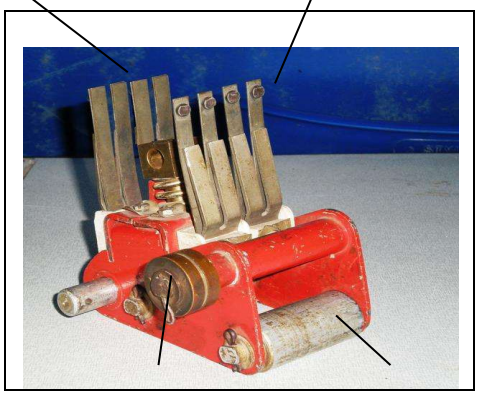


Fig.- Top Roller, Bottom Roller

Sequence of Point Operation :

1. Breaking of Detection Contacts
2. Unlocking of Point
3. Operation of Point in desired position (Normal/Reverse)
4. Locking of Point
5. Make of Detection Contacts

Position of Different Contacts during Normal / Reverse Setting of Points:

Point Status	Normal Control Contacts **(NC)**	Reverse Control Contacts**(RC)**	Normal Detection Contacts**(ND)**	Reverse Detection Contacts**(RD)**
Set & Locked in **Normal Position**	*Break*	*Make*	*Make*	*Break*
Set & Locked in **Reverse Position**	*Make*	*Break*	*Break*	*Break*
Unlocked Point	*Make*	*Make*	*Break*	*Break*

6. DC Motor :

- DC Series split field Motor is used in EPM
- Series motor is used to meet high initial torque &
- Split field so that motor can be rotated in both the direction (Clock wise & Anti-clock wise to Operate the point in Normal/reverse Position)



Fig.- DC Motor for EPM

- **DC Motor for EPM** Contains four field windings
- Two windings are connected in series to form a set of winding.
- Use carbon brushes

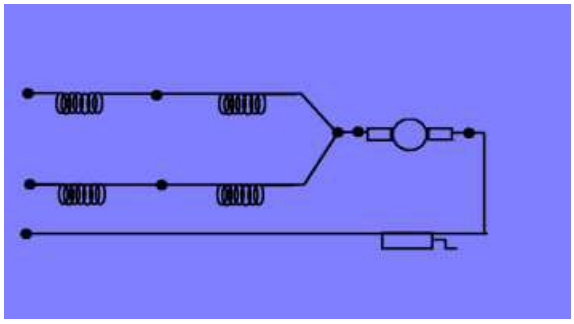




Fig.- Internal windings of DC Motor for EPM

- Carbon brushes are rectangular carbon block which provides the electric connectivity between field windings and rotating commutator
- These are spring loaded in holder with Spring force (4.0-6.0 PSI)

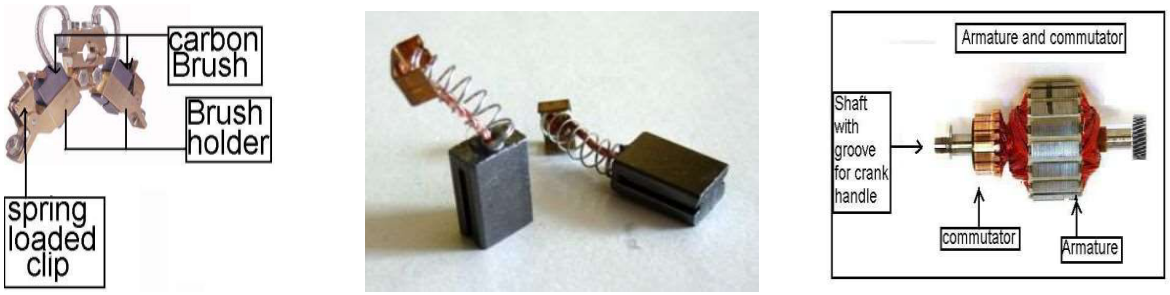


Fig.- Carbon brushes used in DC Motor

Operating Parameters of Motor for EPM :

Rated Voltage – 110V DC (It shall also be capable of operating the Point Machine as per IRS: S 24 between the limits of 75% and 125% of the rated voltage.)
AC Immunity level - 160 V (50Hz) or 400 Volt (50Hz) as mentioned by manufacturer **.**
Insulation Resistance - > 10 Mega Ohm measured with 500V DC Meggar
Power - 0.44 KW
Insulation Type –Class "B"

IP-67 Protection Compliant Motor :

*(**Ref.:** Specification of Motor of EPM no. IRS: S 37-2022 Version 2.0)*

- The Motor shall be constructed to provide IP-67 features as per IS/IEC 60034-5:2000 protocol as to prevent any entry of dust and moisture ingress. Protocol shall be maintained as per international standards which define the level of protection from water and dust.
- The motor shall be capable of functioning normally even after remaining in submerged condition for 24 hours in case of IP-67 Compliant Point Motor.
- IP -67 level shall be clearly mentioned by the manufactured on the motor.

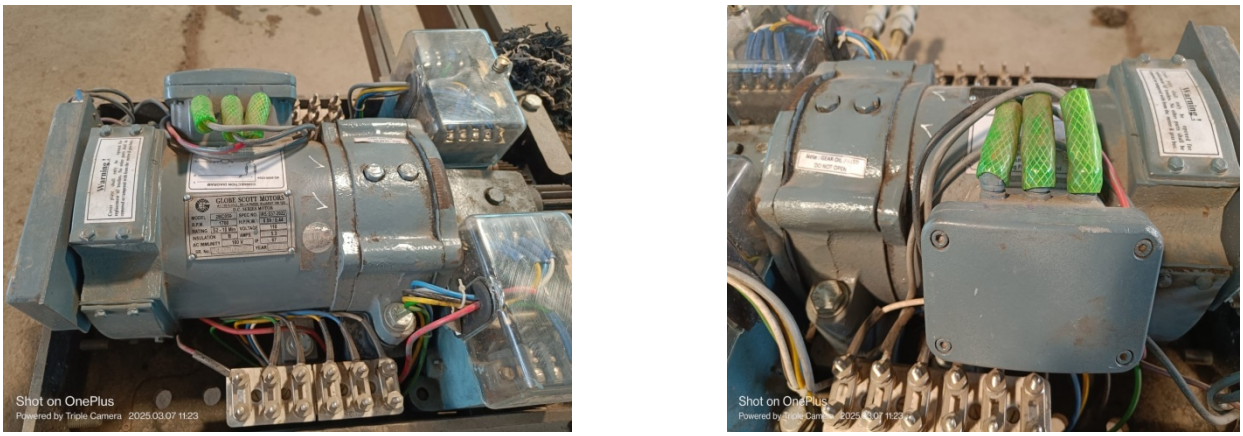


Fig. IP-67 Protection compliant DC Motor for EPM



Fig. Normal (Non-IP 67 Protection Compliant) Motor for EPM

Wiring Diagram of Motor :

*(**Ref.:** Signal Workshop, N.E. Railway ,Gorakhpur)*

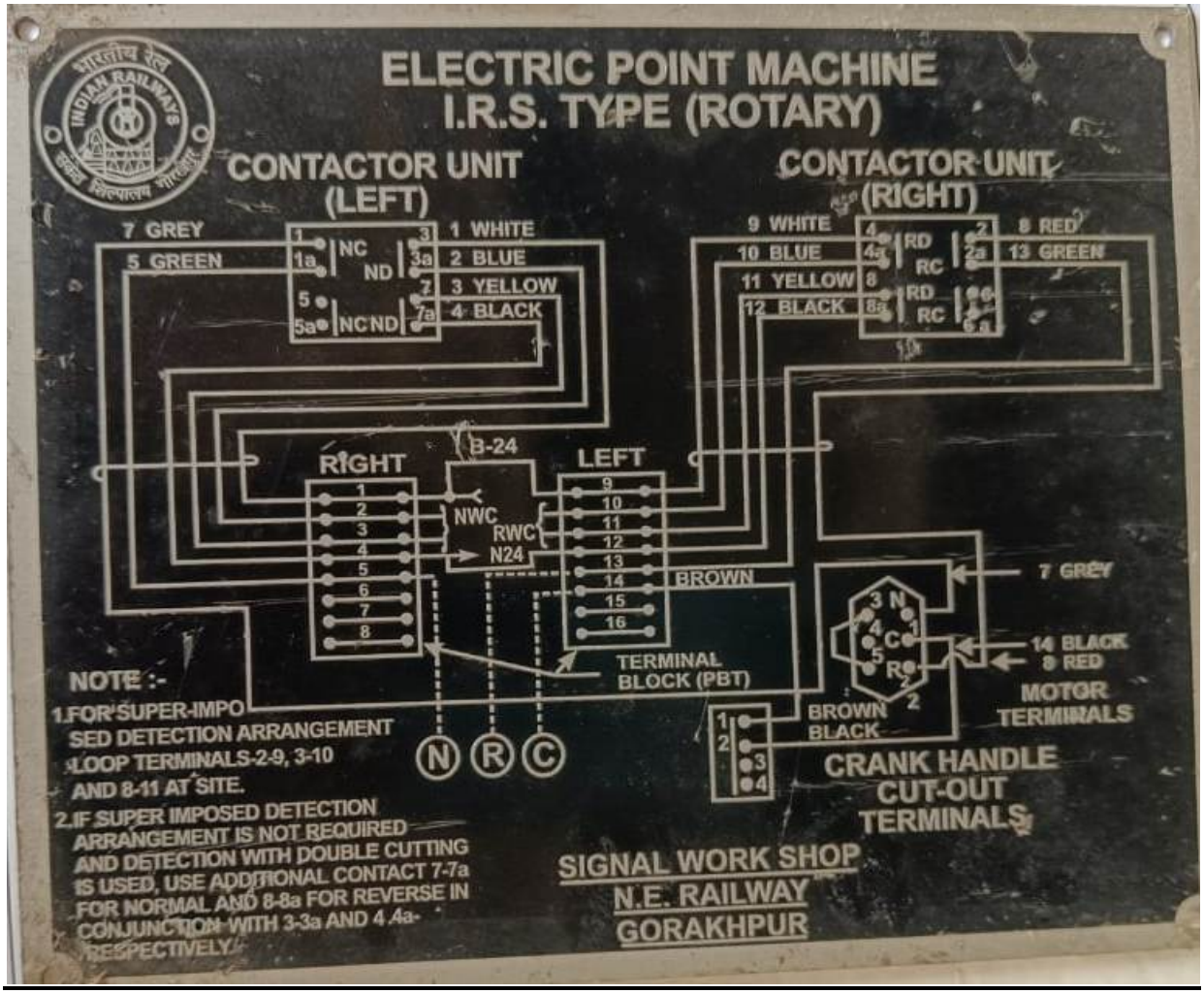


Fig. - Wiring Diagram of Motor for EPM

8. Crank Handle Key :

Insertion of Crank Handle Key -

- Displace the flap cover plate to allow insertion of crank handle
- Open crank handle cut out contacts.

Removal of the crank handle key-

- Replaces flap cover in front of motor shaft to prevent crank handle insertion
- Makes the crank handle contact.



Fig. - Crank Handle Key



Fig. - Crank Handle Key displacing Flap cover and allowing crank handle insertion

9. Crank Handle Cut out assembly:

- There are two sets of contacts in Crank Handle cut out assembly.
- These contacts are wired in negative path of power supply to motor.
- Contacts break / open with insertion and turning of crank handle key &Makes when crank handle key is taken out.
- Crank Handle Key controls the entry of crank handle in the machine for manual operation by moving metal flaps attached in front of motor shaft at the entry point of crank handle.
- Flap cover plate is displaced/replaced by means of a crank handle key



Fig. : Crank handle contact closed & Flap moved in front of shaft



Fig. : Crank handle contact open & Flap displaced & shaft visible



Fig.- **Crank Handle Key operation**

1. 4 GROUND CONNECTIONS :

Ground connections provide rigid moveable connection between the Point machine (Throw bar, detection Slide and lock slides) and the switch (tongue rail /Stock rail) -

The ground connections comprises of the following :

(i) Switch extension brackets (P brackets) – 2 Nos.

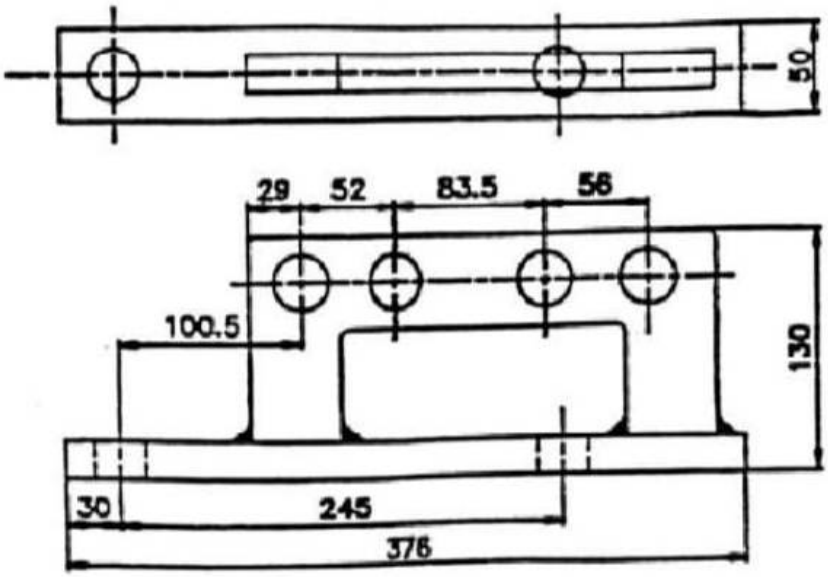


Fig. Switch extension bracket front and top view

(ii) Operating or throw rod.



Fig.. Throw rod (Top view)

(iii) Lock rod long (far end) – 1 No.

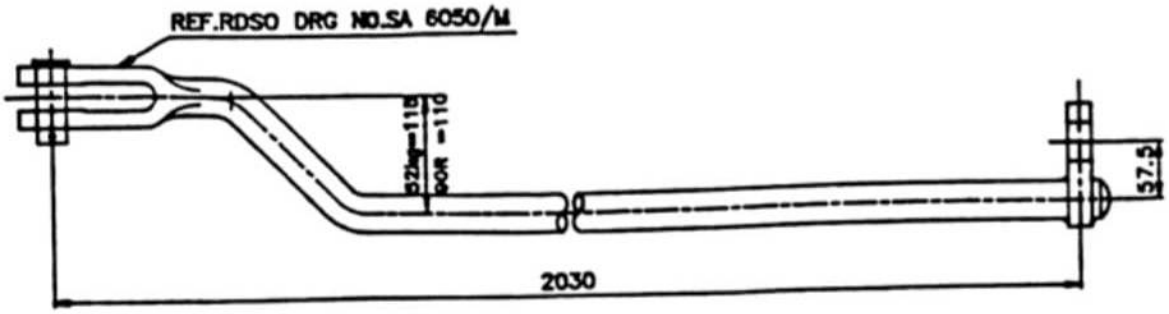


Fig. Lock rod far end (Top view)

(iv) Lock rod short (near end) – 1 No.

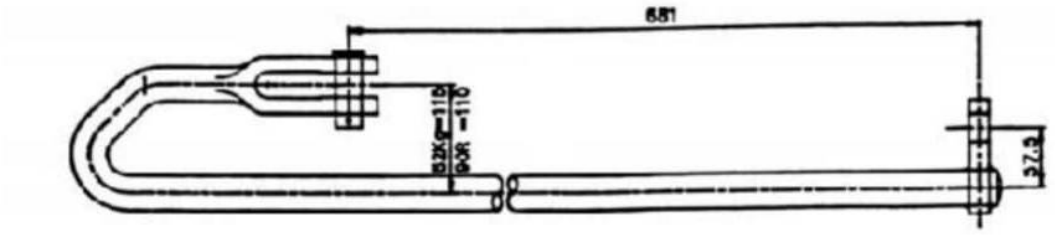


Fig Lock rod near end (Side view)

(v) Detector rod long (far end) - 1 No.

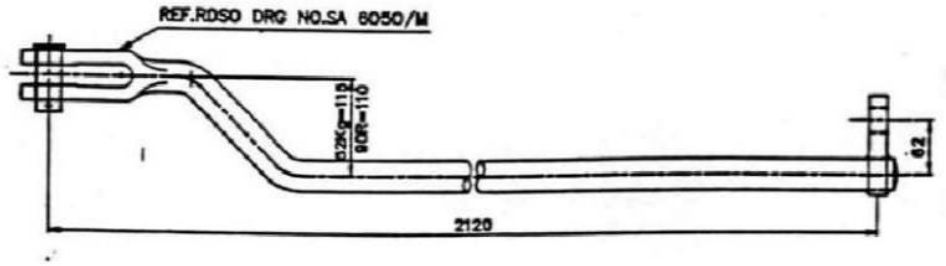


Fig. Detector rod far end (Side view)

(vi) Detector rod short (near end) - 1 No.

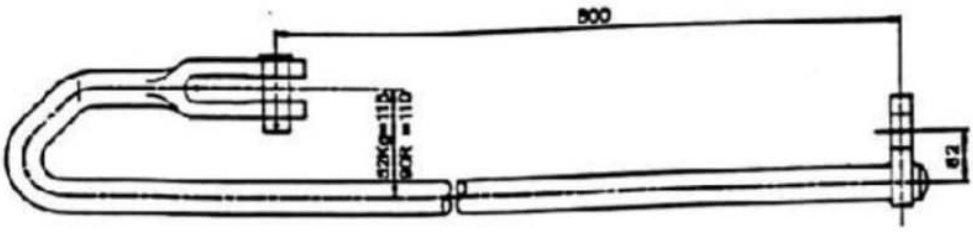


Fig. Detector rod near end (Side view)

(vii) Drive lug – 1 No.

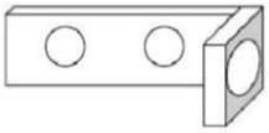


Fig.. . Drive Lug

1. 5 LIST OF INSULATION MATERIAL FOR POINT TURNOUTS IN TRACK CIRCUIT AREA :

1.	**William stretcher bars**	
	Nylon Backing plates	2 Nos.
	Nylon bushes for bolts	3 Nos.
	Nylon washers for bolts & nuts	6 Nos.
2.	**Gauge tie plates**	
	Nylon end post with 3 holes	1 Nos.
	Nylon bushes for bolts	3 Nos.
	Nylon washers for bolts & nuts	6 Nos.
3.	**Electrical detectors provided on mechanical points**	
	Between Detector rods and drop links	1 bush & 2 washers per rod
4.	**IRS type & Siemens point machine** ***(Between tongue rail and tongue attachment (D bracket))***	
	Nylon insulation plate RH (RDSO S3265)	1 No.
	Nylon insulation plate LH (RDSO S3266)	1 No.
	Nylon Bushes for bolts (RDSO S 23199)	2 Nos. Per D bracket
	Nylon washers for nuts (RDSO S 8640)	2 Nos. Per D bracket
	(Between 1st William stretcher & Drive lug)	
	Insulating bush (RDSO S 8813)	2 Nos.
	Insulating plate (RDSO S 8804)	1 No.
5.	**220 mm throw point machine with clamp lock**	
	(Between tongue rail and tongue attachment (P bracket))	

	Nylon insulation plate RH (RDSO S3592)	1 No.
	Nylon insulation plate LH (RDSO S3593)	1 No.
	Nylon bush between lock /detector slide & lock /detector rod lug (RDSO S 23199)	1 No. per detector/lock rod
	Nylon washer between lock /detector slide & lock /detector rod lug (RDSO S8640)	2 Nos. per detector/lock rod
	Nylon bush between drive bar and drive rod lug (RDSO S 3463)	1 No.
6.	**Rodding passing across the track likely to short the two track rails**	
	Liner plate	1 No.
	Bushes for bolts	2 Nos.
	Washers for bolts & nuts	4 Nos.

In addition to above, for 220 mm throw point machine with clamp lock it is to be ensured that
(i) Spring setting device (SSD) is Insulated from both the rails.
(ii) Clamp lock assembly is insulated from both the rails

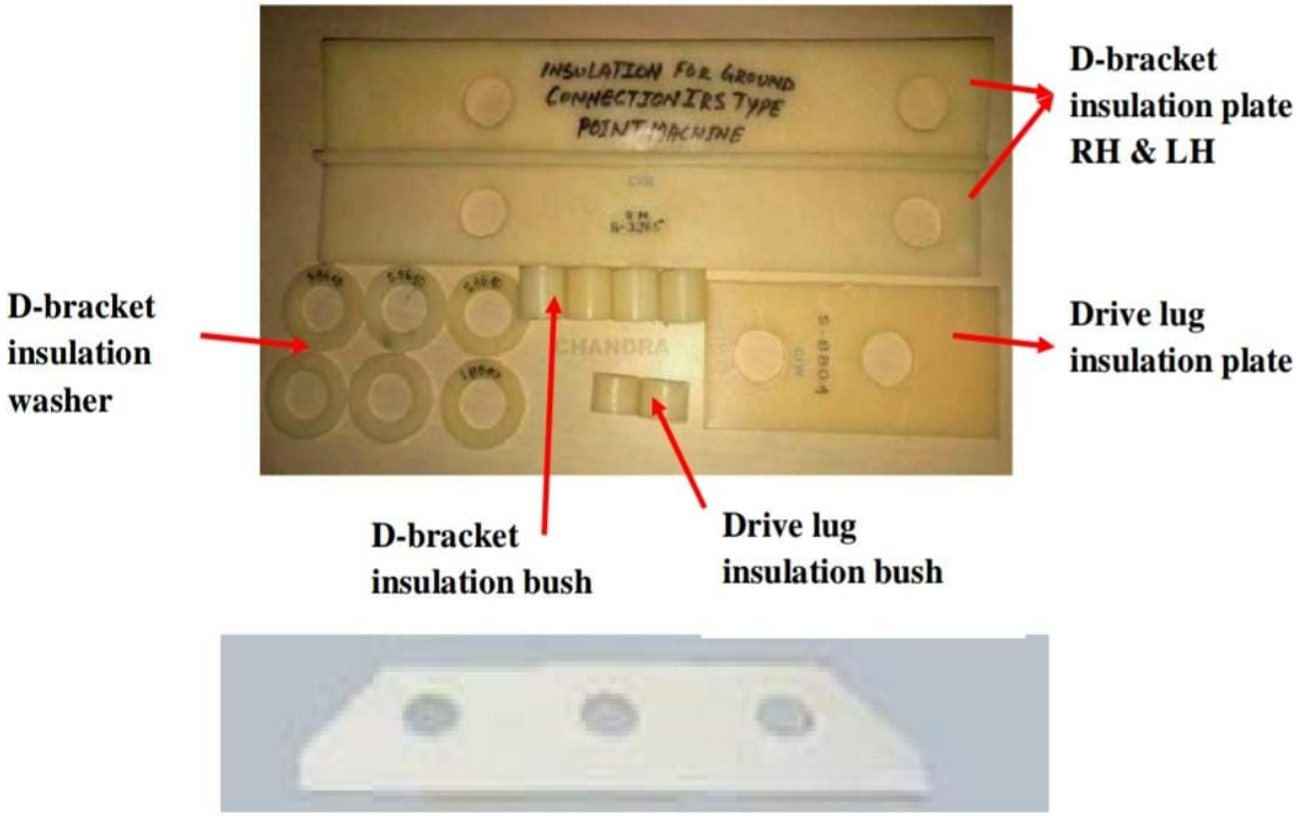


Fig. Insulation material for point turnout

****

CHAPTER-2

INSTALLATION OF ELECTRIC POINT MACHINE

2. 1 INSTALLATION OF ELECTRIC POINT MACHINE (143mm):

(Ref. IRSEM Ver.3.0)

2.1.1. Requirements: Electric Point Machines of approved type shall be used.

(a) Electric Point Machines shall be installed in accordance with approved plans. Ground fittings of approved design shall be used. Ground connections shall be thoroughly checked for any crack or improper welded joint. Normally the machine shall be installed beside the close switch leading to high-speed movements clear of all infringements. Before installation of the machine, it shall be ensured that.

(b) The JE/SSE (P.Way) has made all the provisions at the Points as detailed in para 12.13.2 to 12.13.4 & Annexure: 12-A1 of Chapter 12 of SEM.

(c) The machine has been properly cleaned and lubricated in accordance with the manufacturer's instructions and is working freely.

(d) Proper alignment and level of machine and the point gears.

(e) The top of sleepers which are to support the point machine are at level and that surrounding ballast is well tamped to prevent sinking during the passage of train.

(f) The machine shall be so installed that it is clear of all infringements.

(g) Proper drainage shall be provided to avoid water logging near point area.

(h) Wiring from the point of operation to the location box/apparatus case nearest to the machine shall be in cable. The cable shall be of appropriate size to keep the voltage drop minimum. The connection between location box/apparatus case and point machine shall be through underground cable adequately protected below track and near point machine, through GI/RCC/DWC-HDPE pipe of approved type or any other approved means. It shall be tested for insulation and continuity before bringing into operation.

(i) Crossovers shall be operated by separate point machines, one at each end. The detection of setting and locking of the points at the two ends of a crossover shall be connected in series.

(j) Means shall be provided to cut off the motor feed in case of obstruction to the point movement after a predetermined time lag, based on the type of point machine used. The point operating control circuits shall have track circuit control, overload protection and cross protection.

(k) The crank handles shall be provided to facilitate operation of points in case of failure of point machines. For this purpose, the points in the yard may be divided into convenient groups and to distinguish particular groups, crank handles with different wards shall be used. The slots provided, in the point machines for insertion of the crank handles shall be suitably modified to take only the crank handle applicable to the group. The crank handles shall, however, normally be locked. It shall be possible to release them for use in point machine by releasing a control from the control panel. With the crank handle released, it shall not be possible to clear the signal for the concerned routes. The interlocking of crank handle can also be achieved by using NX key of different wards.

(l) The excess gap around cable entry to the machine shall be plugged to avoid rodent entry/water ingress.

2.1.2 Point Machine, Notches, Gaskets, Sealing:

(a) Electric Point Machine shall be provided with suitable gaskets and wire entrances shall be sealed to make them water tight and dust proof. All the extra openings shall be closed as far as possible.
(b) **Locking:** Electric Point Machine covers shall be locked. The cover provided for insertion of crank handle shall also be locked with separate key providing access only for crank handle and crank handle key.
(c) **Notches in Stretcher Blades:** Notches/Notch profile of the locking, driving and detection slides should not be tampered/reconditioned. Worn-out slides should be replaced with new ones.

INITIAL ADJUSTMENT OF POINT MACHINE

2.1.3 Adjustment of Driving Rod

The points shall be adjusted by operating the machine first by hand cranking. The insertion of hand crank should disconnect the power supply to the machine. When the machine has been fastened down, the throw bar connections shall be set up and the point machine hand cranked to one end of the stroke positioning the locking blades so as to allow the appropriate locking dog to pass through notches. The closed switch shall be adjusted to just in contact with the stock rail. Then the connections shall be tightened further by 2 mm to 3.5 mm to impart a springing action to the tongue rail. The machine should be hand cranked to the opposite end of the stroke and the setting repeated for the other switch.

Note: *It is recommended that adjustment for driving rod and lock rod shall be done for the close switch nearest to the point machine first.*

2.1.4 Adjustment of Detector Contacts:

The machine shall be hand cranked to the end of the stroke to close the tongue rail. Insert 1.6 mm test piece between stock rail and switch rail at 150 mm from toe of the switch and operate the point. Ensure detector contacts just make. A test gauge of 3.25 mm shall be inserted between the switch rail and stock rail at a distance of 150 mm from the toe of the switch. The detector connection of the closed switch shall be adjusted till the appropriate detector contacts are just broken. The same shall be repeated at the other end of the stroke. All the relevant nuts shall be tightened.

Note: *Where lock slide is provided in the machine, during switch detector adjustment the test shall first be done with fictitious locking. After completing the above test, same test be repeated with machine properly locked.*

2.1.5 Adjustment of Friction Clutch:

(a) When IRS rotary type electric point machine under obstruction test draws less than 1.75 times the normal operating current, machine needs to be checked for proper functioning at site.

(b) Obstruction Test: The point driving rod and the lock connections of the machine must be so adjusted that with 5 mm thick test piece obstruction placed between the switch and the stock rail at 150 mm from the toe of the switch:
(i) The point cannot be locked.
(ii) The point detector contacts should not assume the position indicating point closure and
(iii) Friction clutch should slip.

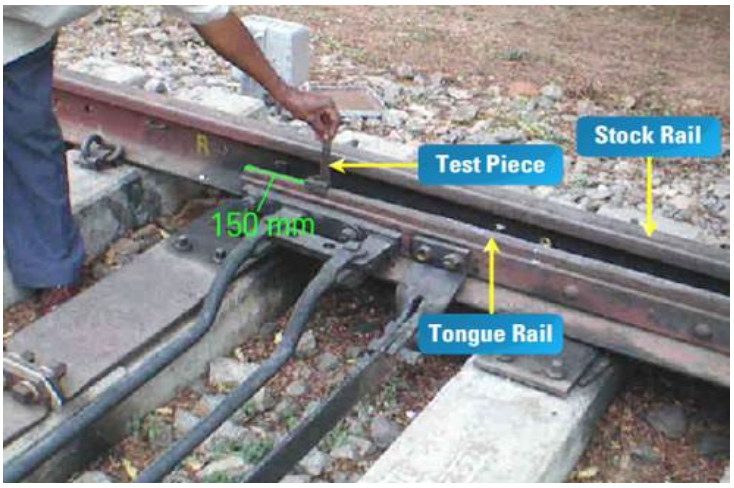


Fig. Obstruction test with Test piece

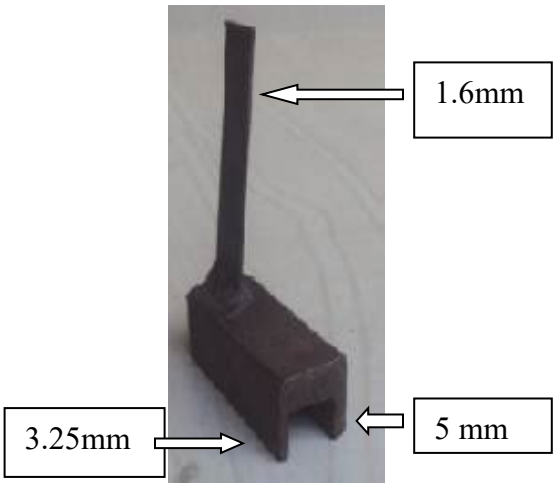


Fig. Test piece

2.1.6 Testing of Point Machine :→

(a) Obstruction Test: The point driving rod and the lock connections of the machine must be so adjusted that with 5mm thick test piece obstruction placed between the switch and the stock rail at 150 mm from the toe of the switch:
(i) The point cannot be locked
(ii) The point detector contacts should not assume the position indicating point closure
(iii) Friction clutch should slip

(b) The Technician shall check whether the current required to operate the machine in either direction is the same (approximately). The reason for unbalance, if any, shall be investigated and eliminated.

Note: *In general, excessive current indicates the friction due to rubbing of rods with rods/sleepers, poor packing, lifting of tongue, improper lubrication, dirt/dust accumulation etc. causing extra load on the machine. The cause to be identified and rectified duly advising Traffic/P.Way department.*

(c)Test the spring in each switch and ensure sufficient spring. It shall not be too excessive to cause excessive wear on machine part, neither it shall too less to hold the switch against the stock rail. In either of the case, the point drive rod shall be adjusted.

(d) Friction clutch where provided shall slip at a value prescribed by the Manufacturer.

(e) Badly worn-out pins shall be replaced; the holes shall be reamed out and fitted with a oversized pin. Split pins where provided shall be properly split out.

(f) Voltage and current shall be checked periodically at the motor terminals, as per Annexure: 19-MS2. This will indicate any undue friction on the points or improperly fastened terminals at cable terminations.

(g) Cable and wire insulation and continuity tests shall be done.

(h) The essential requirement for interlocking of point shall be periodically checked.

(i) It shall be ensured that during obstruction in the point, the feed to point is automatically disconnected or friction clutch is tripped on overload described by manufacturer and reconnects only when a fresh operation is done. Where sequential operation of point machine is done, the maximum time before disconnection may not exceed to 3.5 times the normal operating time of one machine.

(j) Each sectional JE/SSE (Signal) shall test the operating values of the machine and adjustments of the machine by obstruction test every month. The Sectional in-charge of the section shall carry out these tests once in three months.

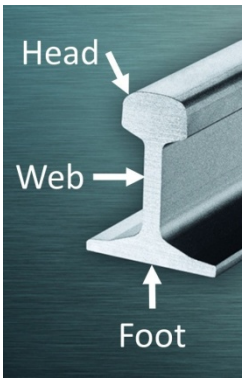
2.2 ELECTRIC POINT MACHINE 220 mm THROW WITH CLAMP TYPE LOCK FOR THICK WEB SWITCHES (TWS):

The traffic density is increasing day by day on Indian Railways and to increase the section capacity higher speeds of trains are always aimed at. To meet this demand, higher power locomotives have been developed and high speed turnouts are being laid. Increased Speed limits on Indian railway have lead to use of heavier 60Kg rails section with Thick Web Switch.

The Conventional 143mm stroke EPM proves inept for operation of these Thick Web Switch, hence IRS Point machine 220mm with clamp type lock has been developed.



EPM 220mm with Clamp type lock



Rail Section

2.2.1 Conventional over-riding type of switches suffers from geometrical deficiencies like:

a) Weakness of tongue rail due to severe reduction of section by heavy machining. Tongue rails are further weakened by heavy wear and tear by traffic. So these are to be replaced frequently.

b) Obstruction to the wheel flange at the actual toe of the switch due to thickness of the tongue rail as well as due to switch entry angle. Hence a jerk is experienced by vehicles on account of these obstructions.

c) Twist in the track over a short distance on tongue rails machined from rail of same cross section, due to overriding nature of the tongue rail.
d) Structural weakness of the entire switch assembly to withstand lateral forces at higher speeds.

2.2.2 Benefits of Thick Web Switches (TWS) over conventional over-riding type of Switches:

Thick Web Switches (TWS) overcome deficiencies of conventional over-riding type of Switches. These are non-over riding type :

a) Thick web switch has wider switch opening of 160 mm (as against of 115mm of conventional over riding type of switches).
b) Ground connection along with clamp lock used for operation of TWS and direct locking of tongue rail with stock rail shall be of approved type.
c) For operation and clamp locking of TWS, electric point machine of approved type with requisite throw shall be used.

2.2.3 For ground connection and clamp lock as per RDSO Drg. No. RDSO/S-3454 for 60 Kg rail and RDSO/S-3455 for 52 Kg rail, IRS point machine of 220 mm throw is used.

a) Out of 220 mm throw, during first 60 mm throw closed switch rail gets unlocked and open switch rail moves towards stock rail. During subsequent 100 mm throw both switch rails move simultaneously thereby open switch gets closed and closed switch gets opened by 100 mm. During final 60 mm throw, closed switch gets locked and open switch rail moves 60 mm further resulting in total opening of open switch by 160 mm.
b) Provides direct locking between tongue rail and stock rail in closed and open position.
c) No stretcher bar is used and both the tongue rails move independent of each other.

2.2.4 Spring Setting Device (SSD):

SSD of approved type shall be provided at junction of Rail head (JOH) to assist in proper setting of switch up to junction of Rail head (JOH).
(a) SSD is placed between sleeper number 13 and 14.
(b) SSD is insulated from both the rails.
(c) SSD installation and maintenance is responsibility of Engineering Department.



Fig. - Spring Setting Device (SSD) with cover open

2.2.5 Ground connection and clamp lock for Thick web switches shall be installed as per concerned RDSO drawings/approved drawings and installation guidelines.

2.2.6 Obstruction Test:

The detector slides, lock slides and drive rod must be so adjusted with 5 mm thick test piece placed between the switch and gauge face of stock rail at 150mm from the toe of the point.
The following to be observed:
(a) Friction clutch de-clutches the motor from mechanism.
(b) The point cannot get locked either by clamp lock or by locking segment inside the point machine.
(c) Lock segment does not enter into the notches of locking slides.
(d) Switch detection contacts do not make.
(e) The slipping current is not exceeding twice of the normal working current.

2.2.7 Insulation Test:

a) Point machine: Check that point machine is insulated from ground connection.
b) Spring setting device: Insulated from both the rails.
c) Clamp lock assembly: Check that it is insulated from both the rails.

2.3 OPERATING PARAMETERS OF EPM (143mm AND 220mm) AS PER IRS: S-24-2025 REV.3. :

(Ref. Specification of EPM No. IRS –S-24- 2025 Rev.3.0)

Throw of point machine	Time of operation (Max.)	Current Consumption of EPM	Current Consumption of EPM with IP 67 Protection	Test voltage (DC)	Test load on throw bar	Slipping load	Stalling load (Min.)
143mm± 2mm	4.0 Sec.	≤4.5 Amp.	**≤5.5 Amp.**	110 V	450 Kg	≤580 Kg	1000Kg
220mm +4/-1mm	5.5 Sec.	≤5.5 Amp.	**≤6.5 Amp.**	110 V	450 Kg	≤580 Kg	1000Kg

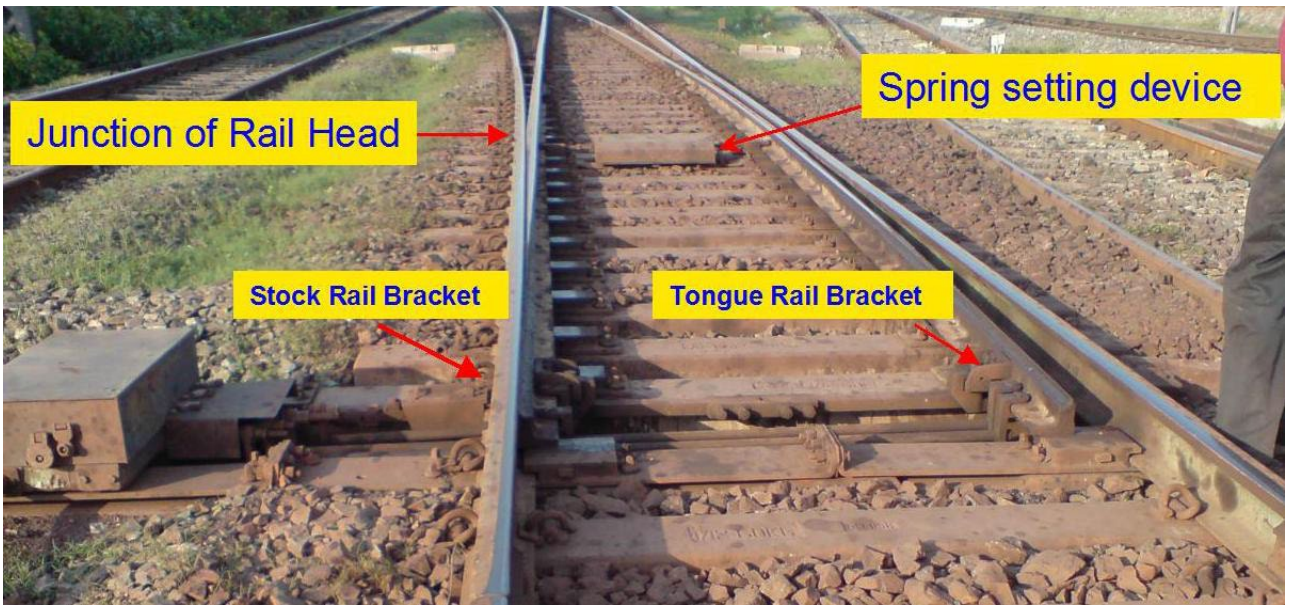


Fig.- IRS-Clamp type Point Machine Layout with TWS

CHAPTER-3

MAINTENANCE OF ELECTRIC POINT MACHINE

3. 1 REQUIREMENTS FOR MAINTENANCE OF POINT AND POINT MACHINE:

Electric Point machine is an electro-mechanical device which can be remotely operated to set the points in required position. For proper working of point machine, prescribed maintenance guidelines should be followed. In addition to this there are certain other factors which may affect the performance of a point machine. Hence maintenance of only point machine is not sufficient; the components which are associated with it directly or indirectly are also to be maintained.

The maintenance of a point and EPM includes the maintenance the following:

(i) Power Supply

Maintenance of Battery Charger/IPS and batteries provided to feed the point motor at site.

(ii) Components of point machine

Dirt and deposit of coal dust in various components of point machine is a source of failure. Hence periodical cleaning of various parts of point machine is required. Wear and tear of carbon brushes, finger contacts in contact pedestal, top and bottom rollers result in failure hence required to be checked frequently. Oiling and greasing of all the moving parts to prevent their wear and tear. Periodical topping up of gear oil in Point motor for smooth operation.

(iii) Electrical parameters

Voltage and current during normal operation and during obstruction as per specified range.

(iv) Cables

Periodical meggering of main and tail cables to minimize cable faults.

(v) Ground connections and fittings

All bolts and nuts including point machine mounting bolts and check nuts are to be checked and adjusted periodically to prevent failure due to point going out of adjustment.

(vi) Insulation in point zone

Insulation materials in track circuited point turnout portion for gauge tie plate and point rodding/ground connections are to be checked periodically to prevent track circuit failure and replaced whenever required**.**

(vii) Safety checks

Following tests are to be done periodically to prevent failure of point or point machine on unsafe side:

- Obstruction test
- Track locking
- Emergency point operation

(viii) P. Way items

In addition to above, the following P. Way items also contribute to the proper functioning of point machine and are to be maintained in co-ordination with Engg. department:

- Lubrication of slide chairs for smooth operation.
- Maintenance of proper gauge and cross level**.**
- Condition of tongue rail
- Condition of stock rail
- Opening of Switches
- Condition Stretcher Bar
- Condition of sleepers and their packing.
- Condition of ballast
- Drainage
- Housing of tongue rail with stock rail

3. 2 PERIODICAL MAINTENANCE OF POINT MACHINE :

Maintenance instructions for point machine and other associated items are given in following sections.

(a) **Cleanliness**: Machines shall be kept in good condition free from rust, dust and dirt.
(b) **Lubrication**: All gearing and bearing shall be properly lubricated according to manufacturer's instructions.
(c) Commutator, Brushes & Electrical Connections
(i) Commutator shall be kept clean, smooth and have bright appearance. Commutator may be cleaned with chamois leather. Under no circumstances, shall emery paper be used.
(ii) Brushes shall be kept clean and properly bedded on the commutator. Brushes shall have proper pressure and shall be free in brush holders.
(iii) All cables connections on terminal, motor terminal etc. shall be checked for intactness.
(iv) All tail cables shall be checked for their insulation as per periodicity given in Annexure: 15-MS1 of SEM.
(d) Adjustment of Contacts: Contacts shall be clean, free from pitting and in proper adjustment.
(e) Rain water drainage arrangements shall be checked especially prior to and during rainy season.
(f) Fixing Nuts & Screws
(i) Fixing nuts and screws of newly installed point machine often tend to get loose. Care shall be taken to attend to them and readjust where required. To prevent this, check nut with spring washer shall be provided.
(ii) All moving parts including locking Dogs and notches slides, etc. shall be checked for chamfering or undue wear. Where badly worn-out, the relevant parts shall be promptly replaced.

3.2.1 Lubrication & Oiling:

- Lubricate all gears and bearings according to manufacturer’s instructions.
- Lubricate the slides, rollers & pins with axle oil Medium grade to IS:1628.Avoid overflow of oil.
- Top up motor gearing with oil as per manufacturer’s instructions through oil inlet

A) Point Machine 143 mm throw, 110 V rotary locking type

(a) Oiling
Motor transmission

Pour 100 cc lubricating oil SAE 30 or Shell 100X through oil inlet (1) (provided on the reduction gear box) into the oil reservoir as shown below. Allow some time for soaking of oil and pour oil in intervals and in quantities of approx-10 ccm at a time until oil emerges out of overflow tube (2). This way excessive overflow of oil is avoided.

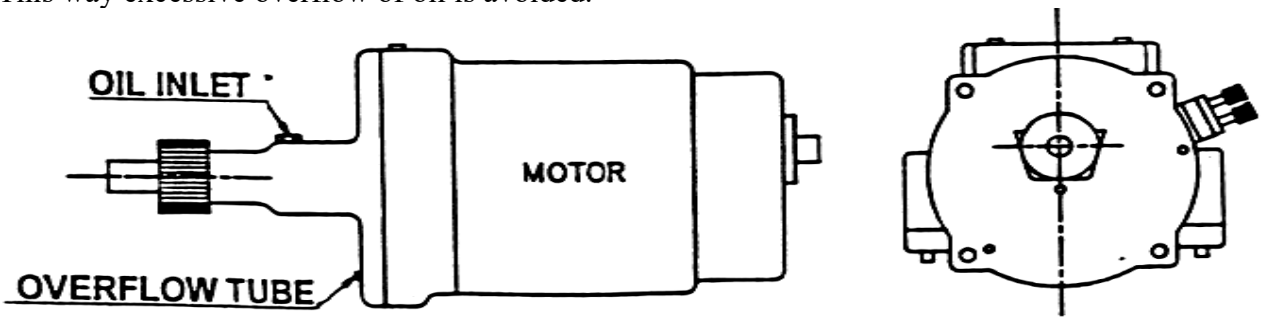


Fig.- Point motor with oil inlet

Periodicity for oiling of motor transmission :

At the time of installation and after 10,000 operations or at 6 month's interval for less frequently operated point machines or as per manufacture's specifications/Zonal Railway instructions. The period may be reduced as and when required according to local climatic conditions.

Throw bar, lock and detection slides
Pour 10 cc lubricating oil SAE 30 or Shell100X in inlets provided on the cover with felt pad for throw bar, lock and detection slides.

Contact assembly
Apply 10 drops of spindle oil on the helical spring guides of contact assembly.

Miscellaneous
Apply medium grade lubricating oil on all pin connections, detection slides, chair plates, lug and sleeve assembly on drive rod etc.

Drainage of excessive oil
To drain out excessive lubricating oil or water accumulated at the bottom of the point machine casting, unscrew the spring loaded drain outlet and close it afterwards.

(b) Greasing:

Apply non-corrosive all temperature grease through all the grease nipples by a grease gun, once in 6 months or as local conditions need to the following:

- Bearings of the gear rack.
- Locking bars.
- Detection slides
- External gears of the transmission assembly.
- Contact assembly of lever bearing.

After approximately 100,000 operations or with less frequently used points, once

annually, wipe off all superfluous grease from nipples to prevent clogging and lubricate thoroughly all internal and external lubricating points.

(B) IRS Electric Point Machine non- trailable with 220 mm throw for Thick Web Switch

(Ref.: M/s Vossloh Cogifer Signalling India Private Ltd .)

Pour oil – BP Energer SHF 48 or Servo System-32 through the oil inlet as shown in figure above until oil emerges out of over flow tube before taking the machine into operation. Pour oil in intervals and in quantities of 20 Cubic Centimeters at a time. In temperature areas ranging from 20 degrees Celsius and less use oil Aero shell fluid4.Spring guides of contact assembly, oiling to be provided 10 drops of spindle oil.

Periodicity of lubrication:

All external nipples of the machine (approximately 10 strokes of grease gun) for every 10,000 operations or every 6 months, if the frequency is less. All external and internal lubricating points- approximately for 1, 00,000 operations. Spring guides of contact assembly, oiling to be provided 10 drops of spindle oil.
Lubricate at the following moving parts of the Clamp Lock fortnightly:
(a) Stock rail bracket groove.
(b) Moving Part of tongue rail & lock arm assembly.
(c) Between matching of lock bar & lock arm assembly.

Ballasting, Packing & Housing:

i) Check for proper ballasting & packing of sleepers - Ballast size > 50 mm
- Clearance below rail bottom –50 mm

ii) Housing of switch with stock rail.
(a) For 143 mm stroke point machine– not less than 11 sleepers or upto JOH is desirable.
(b) For 220 mm stroke point machine with clamp lock for TWS – not less than 13 sleepers

3. 3 MAINTENANCE ITEMS TO BE CHECKED DURING JOINT INSPECTION OF POINTS & CROSSINGS WITH ENGG. DEPARTMENT

For smooth operation of points & point machine, the following are to be ensured which are covered in joint inspection of points & crossings:

3.3.1 Engineering items:

(a) Condition of tongue rail-

- Tongue rail houses properly against the Stock rail.
- If tongue rail is chipped , cracked, twisted or bent, get it replaced.



Fig. - A Chipped tongue rail

Tongue rails should not be out of square -

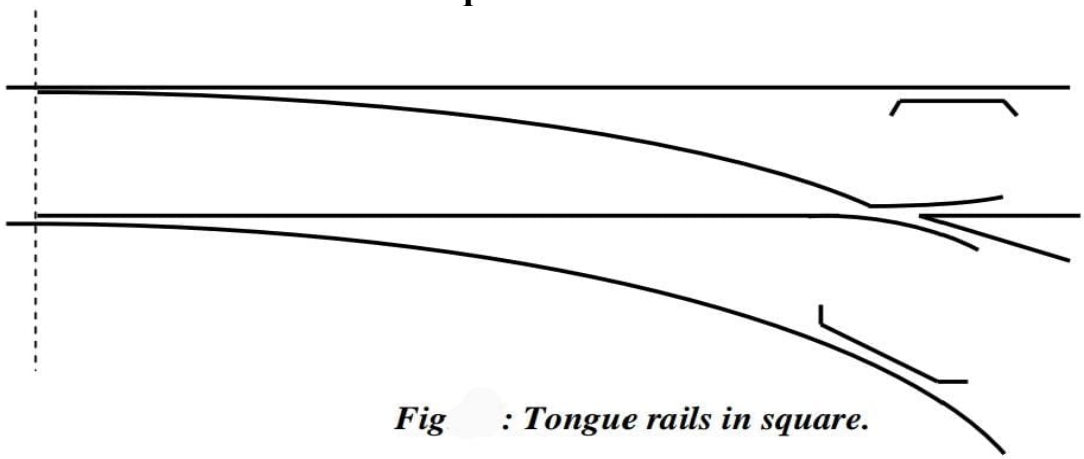


Fig : Tongue rails in square.

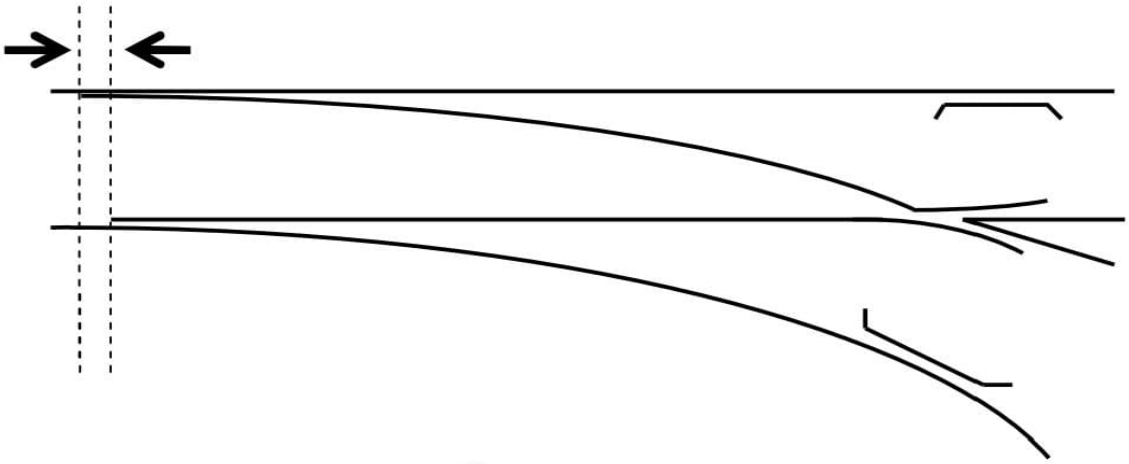


Fig . Tongue rails out of square.

(b) Condition of stock rail

Burrs if any on the stock rail obstructing the housing of switch should be removed. If stock rail still obstructs the housing of switch, the stock rail should be replaced.As far as possible the tongue rail and stock rail should be changed together and not separately to avoid bending of tongue rail due to uneven wear of the rail table.Ensure proper housing of the tongue rail, for both LH and RH switches for the full length of the planning of switch.[317(3) (l) of IRPWM].

(c) Opening of Switches

Minimum clearance between toe of open switch and stock rail
For 143 mm throw point machine - 115 mm
For 220 mm throw point machine with clamp lock for TWS - 160 mm

(d) Stretcher Bar

The clearance between top edge of stretcher bar and bottom of the stock rail shall be at least 1.5mm but not more than 3.0 mm. [Para- 317(3Xk) of IRPWM.Ensure that 1st ,2nd ,3rd and 4th stretcher bars are fitted in proper place. Check whether oblong hole is provided on leading stretcher bar.



Fig.- Clearance between top edge of stretcher bar and bottom of the stock rail

(e) Condition of sleepers

(i) Ensure that the sleepers are not worn out and in good condition:

- Two sleepers below machine.
- Two SRJ sleepers
- Two heal block sleeper

(ii) Spacing between 3rd & 4th sleepers :
For 143 mm throw point machine - 685 mm
(Ref. RDSO drawing no. RDSO/S 3262-63 & SA 9151-52 For 220 mm throw point machine with clamp lock for TWS)

(f) Condition of ballast

Check the condition of ballast and ensure that:

- Ballast size is > 50 mm
- The ballast is without muck
- Clearance below rail bottom is 50 mm

(g) Drainage

- Check if there are any cases of water logging in past.
- Ensure that proper drainage is available for track**.**

(h) Housing of tongue rail with stock rail

Check that the tongue rail houses properly against stock rail.-
(i) For 143 mm stroke point machine – not less than 11 sleepers or upto JOH is desirable.
(ii) For 220 mm stroke point machine with clamp lock for TWS– not less than 13 sleepers or upto JOH is desirable.

(j) Resting of tongue rail on slide chairs

Check that the tongue rail is resting properly on slide chairs - upto 6 sleepers is desirable.

(k) Gauge and cross level

The gauge & cross level shall be measured 150 mm ahead of actual toe of switch.

(l) Packing under switch assembly

Ensure that packing of ballast under switch assembly is proper and there is no vertical movement under train.

(m) Condition of switch fittings

Check that Gauge tie plate, Slide chairs, Stretcher bar, Heal block, Nut and bolts are in good condition.

(n) Spring Setting Device (SSD) for Thick Web Switch

Ensure that :

- Pressure is equal on both sides.
- Lubrication is proper
- Insulation is good

3.3.2 S&T items :

(a) Function of lock detection

(i) Without test piece - Set & Locked

(ii) With 5mm. test piece - Cannot be locked

(b) Time taken in completing the operation

(i) Without test piece - not more than 6 sec.

(ii) With 5mm. test piece - 10 to 15 sec.

(c) Ground connections

Check whether pins snug fit in D bracket and ground connections without hammering.

(d) Testing of Track Locking and Emergency Point operation :

Track locking must be tested quarterly for each track circuit of any point/ cross-over individually. Fail one track circuit with the other track circuit of point/cross-over as pick up.

It must not be possible to operate the point by WN and WWN buttons (Normal operation).

Point can be operated only by pressing buttons WN and EWN (Emergency operation).

Now fail other track circuit with the first track circuit of point /crossover as pick up. It must not be possible to operate the point by WN and WWN buttons. (Normal operation)

In this case also point/cross-over can be operated only by EWN and WN buttons (Emergency operation).

Track locking must be tested with each tack circuit of point/cross-over failed one at a time and other as pick up individually for each and every point/cross-over.

3.4 GENERAL MAINTENANCE ITEMS TO BE CHECKED DURING EVERY MAINTENANCE VISIT –

- All bolts and nuts including point machine mounting bolts and check nuts are tight.
- All pins are tight.
- All split pins are opened correctly.
- Electrical wire connections inside the cable termination box and inside the point machine are tight and the wiring is laced properly. Wires are clear of all moving parts and do not get trapped in the lid
- when closed.
- Slide chair plates are graphited and lubricated for smooth working of points.
- Gear transmission works smoothly without any crackling sound when machine is
- operated by power.
- Brass strip between the detection slides is intact.
- The rollers moving over the friction clutch are not worn out and moving freely.
- Pins of switch extension piece have no rib formation or excessive wear.
- Manual operation with the help of crank handle in both directions is working properly.

3.5 ITEMS TO BE CHECKED BEFORE THE ONSET OF MONSOON:

- Testing of all emergency crossovers for their proper working.
- Checking of insulation of point machines.
- Oiling and greasing of the point machines.
- Availability of gasket in housing cover of point machine.
- Lifting of point machines wherever required at identified water logged area.
- Checking cable entries to ensure that the cable is in a healthy condition physically besides meggering and ELD monitoring.
- Waterproofing of point motors where points are situated in low lying flood prone areas.
- Rodent entry points in the point machine and Cable Termination Box etc. if any are properly plugged.
- Conducting drills for replacement of point motors and detection contact assembly.
- Spare point motors and contact assemblies are kept at stations.

3.6 REPLACEMENT:

As per Specification of Point Machine no. IRS:S24 -2025, the life cycle of Electric Point Machine is 10 lakh operations. However, as per the practice in some railways the point machine should be replaced after -

☐ 7 years - on Sub-urban and Trunk routes, 12 years - on other than Sub-urban and trunk routes

OR

☐ 3 lakh operations whichever is earlier.

3.7 TROUBLESHOOTING (EPM 143 & 220 mm):

Sr. No	Nature of Failure	Possible Cause of failure	Remedy
1.	Point machine does not start.	Fuse blown Off	Replace with new one
		Feed is not coming out from relay room.	Check the circuit and get rectified
		Cable conductor becomes faulty	Megger the conductor, replace if faulty.
		Motor winding got low insulation.	Replace the motor.
		While starting control contacts are not making.	Adjust the control contact
2.	Point is showing gap and motor rotates continuously	Obstruction in between stock and tongue rail.	Remove the obstruction.
		In friction due to improper cleaning and oiling of points.	Clean and check oiling of point.
		Wrong adjustment of point throw rod.	Adjust point through rod.
		Defects with P-Way fitting.	Attend with P.Way staff.
3.	Point is properly set, but motor continuously rotates.	Respective detection contacts are not making (Point may lock)	Adjust detection contact.
		Adjustment required in lock rod, if point is not locked.	Adjust lock rod for proper locking.
		The respective detection terminal loose or broken in point machine.	Attend respective detection contact.
4.	First point is correctly set, locked and detection contacts are making, but feed not extended to IInd point machine.	Feed not coming to IInd point machine due to cable between the machine defective.	Check the cable conductor between point machine, replace if defective or low insulation.
		Crank handle contact of IInd point machine not making.	Adjust crank handle contact.
		Respective terminal loose or broken in IInd point machine.	Tighten terminal or replace broken terminal.
5.	While operating the points the	Any wire	Check wiring and

	motor rotates in just reverse direction.	interchanged.	rectified.
		In point junction box cable interchanged.	Check wiring at junction box and rectified.
6.	First point is correctly set, locked and detection made but the motor do not stop even after starting of IInd point machine.	Short circuit faults in the terminals in point junction box.	Check wiring at junction box and rectified the short circuit faults.
7.	Out of correspondence between point and relay group.	Cable inters changed in point junction box.	Check and rectified.
		Wiring inters changed in crank handle box terminal.	Check and rectified.
8.	Tongue rail not housing up to 13thsleepers or tongue rail housing one side properly but not housing another side at adequate distance.	Spring setting device not adjusted properly	Arrange P-Way staff for proper adjustment of spring setting device.
9.	Clamp lock not locking point on either side.	Packing shims between stock rail and tongue rail bracket or outside the tongue rail are to be adjusted.	Adjust the numbers of packing shims of tongue rail bracket.

3.8 Do's & Don'ts :

Do's.

- Get disconnection of point, before carrying out any work.
- Always ensure schedule maintenance.
- Test the working of friction clutch.
- Do ensure that track locking, route locking, crank handle release locking are effective.
- Do test the correspondence of the points with indication at panel or cabin.
- Always use proper lubricant. Wipe away excessive oil or grease.
- Always use proper rating of fuse.

Don'ts.

- Do not operate the point manually except through specific crank handle.
- Forget to exercise safety checks initially, during and after disconnection.
- Forget to carry necessary tools while attending the failure.
- Leave any discrepancies noticed in the point machine.
- Use too much lubricant.
- Allow unsquared notches on slides.

3.9 TOOLS & SPARES:

Recommended Tools and measuring instruments:

Following tools and measuring instruments should be kept in the duty room for installation and maintenance of Electric point machine:

- Spanner set
- 32 mm (11/4") single end spanner
- Adjustable pipe wrench
- Screw driver
- Hammer
- Tommy bar
- Measuring tape
- Test gauge
- Cutting pliers
- Wire cutter
- Insulation peeler
- DC Ammeter with centre '0' incase of Analog (0-30 Amp)
- DC Voltmeter (0-250V)
- Track shunt resistance.
- Grease Gun

Spares and Consumables

Following spares and consumables are necessary for maintenance and inspection.

- SAE 30/SHELL 100 oil
- Grease IS 507/508
- Chamois leather
- Cotton waste grade A
- Carbon brush
- Contact assembly
- MS-pins and split pins
- D-bracket and stretcher bar insulation material sets

Tools and accessories to be supplied with new point machines:

(Ref. IRS:S24 -2025,)

- The following set of tools in a suitable box shall be supplied along with every set of eight point machines or less:
  1. Hand operated square drive socket wrenches of 12.5 mm driving square (short type) as per IS: 7381 for M8, M10, M12 & M18 threads. (One 250 mm long 12.5 mm square extension bar as per IS: 7991 along with compatible Tee handle Square drive as per IS: 7975 suitable for above socket wrenches.
  2. Open jaw wrenches (spanners) for M10, M12, M20, M24 & M33 threads as per IS: 2028.
  3. Adjustable wrench.
  4. Screw driver 300 mm long.
  5. Additional tools required for **Electric Point Machine with IP67 Protection** for maintenance may be added as per requirements.

- Junction box and two Nos. of telescopic pipes of approved type shall be supplied as an integral part of the with EPM as integral part of Point machine.
- Ground Connections and clamp lock shall be supplied along with point machine on requirement from the purchaser as per latest Drawings given below or as per their requirements .-

  i) RDSO/S 3361-62 (60 Kg ,52 Kg on PSC Sleepers)

  ii) RDSO/S 3262-63 (52 Kg ,90R on Wooden Sleepers)

  iii) RDSO/S 3454 (60 Kg Switch Clamp Point Lock)

  iv) RDSO/S 3495 (60 Kg Swing nose crossing with clamp point lock)

****

CHAPTER-4

CHECK LISTS FOR MAINTENANCE OF ELECTRIC POINT MACHINE

4. 1 CHECKLIST FOR TWS (THICK WEB SWITCH) INSERTION :

Ref. *Sr. DSTE(Co.)/MMCT Western Rly Letter No. SG LAO/2/Sr. DSTE Instruction* ***Date :*** *05.04.2023*

CHECKLIST FOR TWS INSERTION

S&T Items:

1) Modified GC as per latest approved drawing only to be provided-
   (i) Fixing of Point Machine: At a distance of 970 mm from Outer edge of machine to inner edge of stock rail
   (ii) Far end Detection rod-2100mm
   (iii) Far end Locking Rod-2040 mm
   (iv) Near end detection Rod-880mm
   (v) near end locking rod-760mm
   (vi) Down lug of drawing bar-62mm(Reduced from 67mm)
   (vii) Tongue rail hole-markings-
      (a) First hole of tongue rail at 40mm for 'P' bracket.
      (b) Second hole of tongue rail at 285 mm for 'P' bracket
      (c) Third hole of tongue rail at.443mm for clamp lock assembly
      (d) Fourth hole of tongue rail at 513mm for clamp lock assembly
   (viii) Stock rail hole to be marked from toe of tongue rail (ATS)
      (a) First hole at stock rail is 450.mm
      (b) Second hole at stock rail is 530.mm

2) Clamp lock MS Stud safety split pin to be inserted from Sleeper No. 4 side through hole on Clamp Bracket collar.(It prevents outside interference and prevents the possibility of obstruction by split pin ends)

3) Edges of clamp lock covers to be suitably bent outwards to eliminate the possibility of loosening covers, creating obstruction.

4) For 1 in 8.5 TWS points, +20mm driving bars to be used. (By replacing the side plates of driving bar at mid insulated joint.)

5) Additional parallel cable conductors to be added before TWS conversion (TWS draw more current, hence more voltage drop in line. For proper functioning, at least 80 V must be available at motor terminal on obstruction current.)

6) Proper housing of tongue rail against stock rail (with SSD connected) must be ensured before connecting GC. (upto 12th Sleeper)

7) Tongue Rail opening must not be less than 157mm, all efforts must be taken for equal opening on both sides as there are no means to reduce the Tongue Rail Opening, but to equally

distribute it on both sides. (Excess opening has no adverse effect, it merely gives excess room to wheel onthe open Tongue Rail, which is desirable for high speed and also there is a no adjustment available to reduce the same.)

8) Point should be tested for 5mm Obstruction Test. The point should neither be locked nor should it give detection in this. If there is any difficulty to achieve it, it should be corrected jointly as -
   (a) In 5mm gauge testing, when motor stops, Tongue Rail must not open again, if it does so, it shows excessive spring action on Tongue Rail which needs to be corrected.
   (b) Suitable shims may be provided (Max 3x1 mm) to fill such gap between tongue and stock rail to facilitate locking slide adjustment.

9) Freeness of Clamp lock must be checked. No shims should be provided on clamps to reduce the opening of open tongue rail as it reduces the clamp lock assembly tolerance of 3mm which is provided by Design. However for fine adjustment of machine locking slide, few shims (Max 3x1 mm) may be provided.

Engineering Items :

10) MS flats are to be provided as per standard sleeper distances only, sleepers inter distances must be maintained as per RDSO layout starting from SRJ and up to JOH.
   Inter-sleeper Distances-
   (i) Sleeper 3&4 = 745 mm
   (ii) Sleeper 4&5 = 492 mm
   (iii) Beyond 5 up to JOH = 550 mm

   Width & thickness of Machine Plate on 3rd sleeper = 220mm and 14mm
   Width & thickness of Machine Plate on 4th sleeper = 240mm & 14 mm
   (MCI Inserts on Sleeper No. 3 & 4 to be inserted)

11) Square head stopper bolt alignment to be ensured as tilted bolt head or pressed stock rail may create obstruction on sleeper numbers 15, 16 and 17. Hence, proper packing to be ensured on the sleepers.

12) Resting of Tongue Rail on all slide chairs must be ensured with the help of filler gauge when the Tongue Rail is in open condition, corrective action must be taken.

13) SSD squareness and inside lubrication to be checked and following should be ensured:
   a) Close Tongue Rail – Gap should be less than 10mm
   b) Open Tongue Rail – Gap should not be less than 57mm
   c) Closed Tongue Rail – Arm should be at right angle (900)

14) Gauge and squareness of Tongue Rail to be checked at ATS. Creep Marker (i.e. ATS position marking at stock rail with white paint) to be provided to notice for any creep occurred in future, accordingly stock rail clamp position to be adjusted to accommodate it within permissible limit (Stock rail have adjustable notch).

15) All the leaf of the slide chairs must be properly inserted, as loose and half inserted leafs may lead to gauge widening which may badly effect the point machine settings, and may lead to failure.

16) SSD must be connected at linking platform itself, also linking platform must have at least 16 sleepers (presently, it is done on 8 sleepers, which may lead to a problem in laying)

4. 2 CHECKLIST FOR TWS (THICK WEB SWITCH) Maintenance:

Ref. *Sr. DSTE(Co.)/MMCT Western Rly Letter No. SG LAO/2/Sr. DSTE Instruction* ***Date :*** *05.04.2023*

Checklist for TWS Maintenance

1) Proper lubrication of all chairs by grease, starting from ATS to Heal block i.e. up to sleeper No. 14 (First 03 sleepers from ATS by S&T staff and the remaining by Engineering staff.)

2) Point should be tested for 5mm Obstruction Test. The point should neither be locked nor should it give detection in this. If there is any difficulty to achieve it,it should be corrected jointly as -
   (a) In 5mm gauge testing, when motor stops, Tongue Rail must not open again, if it does so, it shows excessive spring action on Tongue Rail which needs to be corrected.
   (b) Suitable shims may be provided (Max 3x1 mm) to fill such gap between tongue and stock rail to facilitate locking slide adjustment.

3) Resting of Tongue Rail on all slide chairs must be ensured with the help of filler gauge when the Tongue Rail is in open condition, corrective action must be taken.
4) Freeness of Clamp lock must be checked. No shims should be provided on clamps to reduce the opening of open tongue rail as it reduces the clamp lock assembly tolerance of 3mm which is provided by Design. However for fine adjustment of machine locking slide, few shims (Max 3x1 mm) may be provided.
5) Tongue Rail opening must not be less than 157mm, all efforts must be taken for equal opening on both sides as there are no means to reduce the Tongue Rail Opening, but to equally distribute it on both sides. (Excess opening has no adverse effect, it merely gives excess room to wheel on the open Tongue Rail, which is desirable for high speed and also there is a no adjustment available to reduce the same.)
6) MS flats to be provided as per standard sleeper distances only, sleepers inter distances must be maintained as per RDSO layout starting from SRJ and up to JOH.
   Inter-sleeper Distances:
   (i) Sleeper 3&4 = 745 mm
   (ii) Sleeper 4&5 = 492 mm
   (iii) Beyond 5 up to JOH = 550 mm
7) Proper housing of close Tongue Rail and smooth gliding on slide chairs must be ensured, any hit marks on Tongue Rail should be attended suitably by the provision of sleeper packing/Tongue rail grinding/rubber padding below stock rails etc.

8) SSD squareness and inside lubrication to be checked and the following should be ensured:
   a) Close Tongue Rail – Gap should be less than 10mm
   b) Open Tongue Rail – Gap should not be less than 57mm
   c) Closed Tongue Rail – Arm should be at right angle (900)

9) Square head stopper bolt alignment to be ensured as a tilted bolt head or pressed stock rail may create obstruction on sleeper numbers 15, 16 and 17. Hence, proper packing to be ensured on the sleepers.

10) Gauge and squareness of Tongue Rail to be checked at ATS and corrective action must be taken immediately. Creep Marker (i.e ATS position marking at stock rail with white paint) to be provided to notice for any creep occurred, accordingly stock rail clamp position to be adjusted to accommodate it within the permissible limit (Stock rail clamp has an adjustable notch).

11) Inside point machine, proper greasing must be done with grease gun by S&T staff. (Recommended grade grease should only to be used).

12) Inside point machine all electrical contact must be properly cleaned, finger contact pressure should be around 3mm and any groves formation in locking slides also to be cleared.

13) All GC split pins to be checked for its integrity and its proper sizes as excessive long pins may create shorting points leading to track circuit failures. (Split pins to be inserted from 4th Sleeper side through hole provided for this purpose).

14) All the leaf of the slide chairs must be properly inserted, as loose and half inserted leafs may lead to gauge widening which may badly effect the point machine settings, and may lead to failure.

15) To curb the gauge variation after any diverging movement as observed, mark-V pendrol clips should be ensured as recommended by competent authorities. It will reduce the frequent point failures for R-N operation after any movement in reverse position besides proper level of stock rail on heel side, that on Sleeper No. 15, 16, 17.

4. 3 MAINTENANCE SCHEDULE OF ELECTRICALLY OPERATED POINTS (INCLUSIVE OF CLAMP TYPE POINT)

(Ref.: ANNEXURE: 19-MS2 OF IRSEM VERSION 3.0)

Schedule Code: EP1			
Periodicity:		*Technician(Signal)*	*Fortnightly*
		Sectional SSE/JE(Signal):	*Monthly*
		SSE (Signal)/Incharge :	*Quarterly*
Sr. No.	**Maintenance Activity (Check the Following)**		
1.	The machine for tightness and free from rust & dirt. Cleaning, graphite/oiling of slide chairs. Lubrication of slide chairs and assembly up to 3 sleepers from the toe of switch by Signal staff.		
2.	Checking of Point Gear Assembly, slides, rollers & pins. Ensure that roller is free from wear and tear and falls freely on control and lift out disc.		
3.	Tightening of all nuts, check nuts & bolts, lock nuts holding the detector slides & lock slides with lugs and condition of split pins to be checked.		
4.	The Lubrication/Greasing of all gears and bearings, cleanliness & smoothness of commutator, checking contacts for freedom from pitting and proper adjustment.		
5.	Visual checks of Points insulations and stretcher bars not rubbing with any fixture.		
6.	The contacts for proper adjustment contact pressure& free from pitting. Wires are neatly dressed & clear of all moving part. Ensure they do not get trapped in the lid when closed.		

7.	All the bridge contacts make & break at the same time.
8.	The setting of switch for having required amount of spring action.
9.	**Obstruction test of points** with 5 mm test piece (to be kept at 150 mm from the toe of the switch) to ensure that point cannot be locked, detection contacts shall not make & friction clutch should also slip. However Detection contacts shall make with obstruction of 1.6 mm test piece (to be kept at 150 mm from the toe of the switch) Testing may be done either by Crank Handle or by Point Motor operation.
10.	Also ensure that both sleepers are well packed & Ground connection rods are free from ballast.
11	Checked the insulation of Gauge tie plate, all stretcher bars, P/D brackets & driving lug and replaced if found damaged/broken.

Schedule Code: EP2

Periodicity	*Technician (Signal)*	*Monthly* *(To be done by Technician (Signal) in the presence of SSE/JE)*
	Sectional SSE / JE(Signal)	*Monthly*
	SSE(Signal / Incharge	*Quarterly*

S.No.	**Maintenance Activity (Check the Following)**
1.	Measurements of operating values (voltage & current) of point machines, with and without obstruction for normal and reverse operation. Current required to operate the machine in either direction shall be 1.5 to 2 times of its normal operation and friction clutch shall slip within this range. Replace machine when difference between normal operating current and current under obstruction is less than 0.5 A.
2.	Checking of feed disconnection time under obstruction is not less than 10 Seconds.
3.	Ensure Hose pipe/GI pipe in good condition and without gaps/access.
4.	Check MS pins of Switch Extension piece/'P' bracket for any rib formation or excessive wear.
5.	In case of Clamp type point machine, Lubricate the following moving parts of the clamp lock. (a) Stock rail bracket groove. (b) Moving part of tongue rail and lock arm assembly. (c) Between machine of lock bar and lock arm assembly

Schedule Code: EP3

Periodicity:	*Sectional SSE/JE(Signal):*	*Quarterly (Sectional SSE/JE & Incharge SSE to carry out alternate inspections)*
	SSE(Signal)/ Incharge:	*Quarterly (Sectional SSE/JE & Incharge SSE to carry out alternate inspection)*

S.No.	Maintenance Activity (Check the Following)
1.	Joint check with JE/SSE (P-Way) of points & crossing for levelling, squaring, creeping, packing, clearance of ballast and other P-Way fittings, etc. and measurement of LH,RH switch opening are as given below for normal point and as per proforma circulated by RDSO dated 14.2.19 for Thick Web Switches.
2.	Joint checking of SSD Setting and its arm insulation with P-Way supervisor.

Normal Point (143 mm) OPENING (Tolerance)		**Thick Web Point (220 mm)** OPENING (Tolerance)	
LH END	RH END	LH END	RH END
115±3 mm	115±3 mm	160±3 mm	160±3 mm

Schedule Code: EP4

Periodicity:	*Technician(Signal)*	*Quarterly*
	Sectional SSE/JE(Signal)	*Half-yearly*
	SSE(Signal)/Incharge	*Yearly*

S.No.	Maintenance Activity (Check the Following)
1.	Greasing/Oiling of point machine and Checking of all grease nipples in position.
2.	Oiling of Point Gear Assembly, slides, rollers & pins with medium grade axle oil IS 1628. Avoid overflowing.
3.	Smoothness& cleaning of Commutator, carbon brushes.
4.	Ensure painting of connecting rods is satisfactory.

Schedule Code: EP5

Periodicity:	*Sectional JSSE/JE(Signal):*	*Half-yearly*
	SSE(Signal)/Incharge :	*Yearly*

S.No.	Maintenance Activity (Check the Following)
1.	Check for detector contacts and cleaning if required, control contacts, friction clutch. Ensure contact pressure of control and detection contact is adequate. Ensure Brass tip on finger contact is intact. Conduct obstruction test.

2.	Visual check of brass strips provided between detector slides, without removing them.
3.	Checking of contact, connections and its effectiveness during power operation points.
4.	Working of point using crank handle shall also be checked. It shall not be possible to insert Crank Handle without assigned Key. Interlocking with signals shall be checked
5.	Checking of point motor insulation, cable and wire insulation (by 100 V Megger)
6.	Testing of point tail cable from K Rack in N & R position of point with 100 V Megger.

Note : *Whenever any wire, cable, gears etc. are opened and disconnected, care should be taken for proper reconnection of wires, cables, gears etc. and must be followed by correspondence test with panel before giving reconnection.*

*****

CHAPTER-5

COMMON FAILURES OF ELECTRIC POINT MACHINE AND GUIDELINES TO MINIMIZE THEM

5.1 COMMON FAILURES IN ELECTRIC POINT MACHINE) -

Some common failures observed in electric point machines in the Indian Railways are given below:

5.1.1 Mechanical Failures :

a) **Friction Clutch Malfunction**:
The friction clutch, designed to protect the motor from overload and ensure smooth operation, can fail due to wear and tear or improper adjustment. This may lead to the point machine stopping mid-operation or failing to complete the switch movement, causing delays or disrupting train movement.

b) **Gear and Transmission Wear**: The reduction gears and transmission assembly, which convert rotary motor motion to linear movement, can wear out over time or fail due to inadequate lubrication, leading to sluggish operation or complete failure.

c) **Top Roller Worn-out:** Due to ingress of dust, over a time rollers have tendency to rub against the periphery of Control and lift out disc. This results in flattening of roller surface and sticking of roller in notch of control/lift out disc and failure of point. Hence periodical cleaning and lubrication of roller assembly is must.



Fig (a):Roller in good condition



Fig. (b): Worn out roller

d) **ROLLERS LIFTING PROBLEM:** After complete operation and setting of point the detection rollers tend to lift out of control disc causing point indication failures due to Friction clutch trouble.

e) **Breakage of Lock slides:** In recent past, incidents of breakage of lock slide have also been reported.



Fig. Images of Breakage of Lock slide

The failure of the reverse lock slide may be attributable to unusual movement of locking slide and lack of lubrication.

(Ref. Dy. CCMT/LGDS S.C.Rly letter No. No.CM 12/3/FI/Carriage Date: 23.02.2022)

Remedial measure: *Sharp change in section may be suitably radiused to avoid stress concentration location.*

Note :
Avoiding installation of point on curve : In initial work point installation on curve may be avoided since it will increase load on the point machine for setting of points and lead to breakage of lock slide.

5.1.2 Electrical failures :

a) **Power Supply Problems**: Voltage drops in cable, Short circuits, loose connections, or interruptions due to poor Earthing can prevent the point machine from operating.

b) **Motor Failure**: The DC series split-field motor, typically operating at 110V DC, can fail due to overheating, insulation breakdown, and carbon brush wear. Poor maintenance or excessive load from obstructed points can exacerbate this issue.

c) **Detection Contact Issues**: Detection contacts, which confirm the correct position of the switch rails, may fail due to dust accumulation, corrosion, misalignment/displacement or low tension in figure springs. This can result in point failure and disrupting train movement.

5.1.3 Other factors :

a) **Obstruction in Point Movement**:
External factors like ballast stones, dirt, debris falling from trains, or even vegetation between the switch and stock rails can obstruct point movement, causing the machine to stall or the friction clutch to slip excessively causing it to malfunction.

b) **Environmental Factors**: India's diverse climate—monsoons, extreme heat, and dust-can degrade components. Water logging during rains can short-circuit electrical parts leading to failure of points.

c) **Engineering Department Factors -** Points failure may occur due to improper packing and less ballast cushion causing loosing packing indication failures. Points may also fail due to improper tightening of all enginnering fittings connected to point like Guage tie plate, Stretcher bar assembly ,Chair plates ,Spikes of point sleepers and creep of point sleepers.

5.2 EFFORTS TO MITIGATE FAILURES OF ELECTRIC POINT MACHINE :

Preventing failures in electric point machines on Indian Railways requires a combination of proactive maintenance, technological upgrades, environmental considerations, and operational best practices.

The failures of Electric point machine can be mitigated with proper design, regular inspection, and maintenance procedures. Availability of spare parts and training of Railway personnel to troubleshoot and repair the machines is also crucial to minimize disruptions.

Below are some strategies to achieve this:

5.2.1 Regular Maintenance and Inspection :

- Conduct routine checks on point machines to ensure all components (e.g., motors, gears, friction clutches, locking mechanisms, and detection contacts) are functioning correctly.
- Lubricate moving parts regularly with appropriate grease and oil (e.g., SAE 30 oil for gearboxes) to reduce wear and tear periodically
- Test the friction clutch periodically to ensure it adjusts properly under load and prevents damage during obstructions.
- Attending joint deficiencies along with Engineering department regularly like point packing , improving ballast cushion, arresting gauge spreading and tightening all other fittings regularly and replacement on condition basis of engineering fittings.
- Periodic replacement of point machine insulations like P-Clamp, Gauge tie plate, Drive lug , Stretcher bar insulations to avoid point zone track thereby avoid point operation failures.
- Replacement of worn-out top rollers and ground connections depending upon condition basis

5.2.2 Condition-Based Monitoring :

- Implement real-time monitoring systems using sensors to track electrical, mechanical, and environmental data (e.g., voltage, current, temperature, and vibration).
- Use predictive maintenance techniques, such as machine learning algorithms, to analyze sensor data and predict potential failures before they occur.

5.2.3 Protection against Environmental Factors:

- **Flood Prevention**: Lifting of point machines wherever required at identified water logged area / flood-prone locations to protect point machines from water ingress, a common cause of failure during monsoons.
- Lifting point machines which are prone to flooding by using **C- type channels and modified ground connections** to suit the lifted points.



Fig. : *Lifting of Point machine with C-Type channel (Rail Support))and modified ground connection*

- **Electric Point Machine with IP 67 Protection** may also be used in such areas along with Galvanized ground connections .



Fig. Images of Galvanized ground connections manufactured by S&T Workshop Podnur,S.Rly.

- Coating of point machine and terminals with epoxy gel and replacement of gasket of cover to avoid water ingress.

- **Debris Management**: Falling debris from trains may be keep out from peripheral of electric point machine and ground connections by frequent cleaning and avoiding outside interference like stone in between stock and tongue rails.

5.2.4 Upgrading Infrastructure :

- Replace outdated point machines with modern designs that comply with Indian Railway specifications (e.g., IRS: S-24/2025), featuring **IP 67 Protection.**
- **Replacement of over aged point machines on no. of operation (300000 lakh operations)**
  Ref : RB letter No. 2015/AIMS&AR/ED/FAR/Ed's Committee dated 18.08.2016

Item Name	**Life in terms of No. of Operations**	**Average Life in years**				
		Routes				
Electric Point machine	**3,00,000**	A	B	C / Suburban	D & Spl.-D	E & Spl.E
		12	12	7	15	15

- **Overhauling Periodicity :**
  (Ref. *PCSTE (Project) / S.C.Rly / Secunderabad letter no. SCR WMFT0SNT(WKSP)/229/2020) dated 17.08.2021***)**

I. Overhauling of point machine is to be done for the machine having operations less than 50 operations per day on condition basis.
II. To implement overhauling schedule, points shall be divided in to four groups based on the frequency of operations as under:-
Group A – More than 100 operations per day - Periodicity of overhauling **-5 years.**
Group B – More than 50 operations per day - Periodicity of overhauling -**10 years.**
Group C – More than 20 operations per day - Periodicity of overhauling -**15 years.**
Group D – Less than 20 operations per day - Periodicity of overhauling - **20 years.**
III. Statistics from data loggers shall be taken to arrive upon the no. of operations of points per day for classification as indicated above.

5.2.5 Power Supply Reliability:

- Install IPS to maintain reliable power supply at all times.
- Replace aging cables and ensure adequate voltage (e.g., 110V DC) to prevent voltage drops that impair performance.
- Providing Spare 110V DC power cable in Ring fashioned manner to avoid voltage drops on the farthest point from the Cabins or Cable huts.
- Providing Fuses with proper rating for point operation and detection circuit like normal(16A) and TWS(20A) points to avoid fuse blown off cases.
- Improving number of conductors for point operation cable from 2 to 4 conductors incase of TWS points (NW,RW,CW).
- By paralleling of Relays contacts used in point operation and detection circuits relays like NWR/RWR,WNKR/WRKR which may avoid relay contact failure cases.
- Carrying out cable meggering of tail cables in every six months and shifting to healthy conductors.
- Frequent cleaning of commutator carbon brushes and replacement as per worn-out basis will avoid point motor failures.
- Frequent checking of insulation of point Machine motor .
- Frequent cleaning of Armature with Chamois leather for avoid high resistance formation.
- Proper wiring of point machines with flexible multi-core coils with suitable lugs and sleeves.
- Proper oiling and cleaning of cut out contact assembly and contacts.

- Proper checking of Point wiring like Track locking and out of correspondence with other points regularly and during replacement and rewiring works.

5.2.6 Learning from Failures:

- Analyze past incidents (e.g., via data from the Signalling Department) to identify common failure points and address them systematically.
- Conduct regular staff training to improve installation, adjustment, and troubleshooting skills.

By integrating these measures, Indian Railways can enhance the reliability of electric point machines, minimize disruptions, and improve overall safety and efficiency.

******

REFERENCES

1. **Indian Railway Signal Engineering Manual (IRSEM) Version-3, July-2021.**
2. **Indian Railways Specification of Electric Point machine No. IRS-S-24/2025 Rev.3.0.**
3. **Indian Railways Specification of Motor for Electric Point machine No. IRS-S-37/2022 Rev.3.0.**
4. **Sr. DSTE(Co.)/MMCT Western Rly. Letter No. SG LAO/2/Sr. DSTE Instruction Date : 05.04.2023**
5. **Dy. CCMT/LGDS S.C.Rly letter No. No.CM 12/3/FI/Carriage Date: 23.02.2022)**
6. **RB letter No. 2015/AIMS&AR/ED/FAR/Ed's Committee dated 18.08.2016.**
7. **PCSTE (Project) / S.C.Rly / Secunderabad letter no. SCR/WMFT0SNT(WKSP)/229/2020) dated 17.08.2021**
8. **Information received from MDZTI/S.C.Rly/Secunderabad**
9. **Information received from Signal Workshop, N.E. Rly /Gorakhpur**

****

QUALITY POLICY

To develop safe, modern and cost effective railway technology complying with statutory and regulatory requirements, through excellence in research, designs and standards by setting quality objectives, commitment to satisfy applicable requirements and continual improvements of the quality management system to cater to growing needs, demand and expectations of passenger and freight traffic on the railways through periodic review of quality management systems to achieve continual improvement and customer appreciation. It is communicated and applied within the organization and making it available to all the relevant interested parties.



INDIAN RAILWAYS
Centre for Advanced Maintenance Technology
Maharajpur, Gwalior (M.P.), Pin Code – 474005